Year Evaluations

Organization. Each deadline has two zero topics for training.
methods expected; they can be given at the beginning of the sequence, even if all the notions
The table shows the expected time of distribution.
and B are reserved for evaluation.

Scales. Zero subjects retain four exercises over five points. Each A/B has two
new 4 and 6 points exercises and two 5 points exercises: total 20.
question by question. For a 1-point question requesting evidence, assign
up to 0.5 points to the relevant procedure and up to 0.5 points to the justification and conclusion.
For an induction noted on 2 points : property and initialization 0.5 ; hypothesis at the fixed rank and
calculation of inheritance 1 ; quantified conclusion 0.5. For another value, adjust these proportions. Do not
apply double sanction to a propagated error. The corrections write the reasonings and the
expected calculations; accept any correct method in the field taught.

Duration. The 2 h DS consists of four short exercises; the 4 h whites consist of four
developed exercises. Adjust only after a real return to the class. The distribution of points
vary between exercises; the total remains 20. Durations are to be adjusted according to the observations in the class.

Due date	Placement benchmark	Zero subjects
SP1	S9 · week of 21 September	From S2
SP2	S18 · week of 12 October	From S14
Common duty	S28 · week of 16 November ; niche to be harmonised	From S14
SP3	S34 · week of 30 November	From S21
White 1	S45 · 14-16 January, 4 h slot to be fixed	From S36
SP4	S57 · week of 8 February	From S44
SP5	S68 · week of 22 March	From S52
White 2	S71 · April 1-3, 4 h slot to be fixed	From S59
Final white	S90 · week of May 31, 4 h slot to be fixed	From S75

DS1 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 1$ and, for the integer natural n, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $1 \leq u_n < 8$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 8$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 9$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+5\vec{\jmath}+9\vec{k}, \quad \vec{z}=4\vec{\imath}+5\vec{\jmath}+10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 5 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS1 · Zero 1

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 4 = \frac{9}{2}, \qquad u_2 = \frac{1}{2} \times \frac{9}{2} + 4 = \frac{25}{4}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 8 - 7\left(\frac{1}{2}\right)^n$.

   Initialization. For $n = 0$, $8 - 7 \times 1 = 1 = u_0$. So $P(0)$ is true.

   Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 4 \\ &= \frac{1}{2}\left[8 - 7\left(\frac{1}{2}\right)^n\right] + 4 \\ &= 4 + 4 - 7\left(\frac{1}{2}\right)^{n+1} \\ &= 8 - 7\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

   It's $P(n+1)$: the property is hereditary.

   Conclusion. The principle of recurrence gives $u_n = 8 - 7\left(\frac{1}{2}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{2} < 1$, for everything $n \in \mathbb{N}$ We have $0. < (\frac{1}{2})^n \leq 1$. Multiplication by $-7 < 0$ reverse inequalities:

$$-7 \leq -7(\frac{1}{2})^n < 0.$$

   By adding 8 and using the previous formula, you get $1 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{2})^{n+1} + 7(\frac{1}{2})^n \\ &= 7(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{7}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

   The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 7(\frac{1}{2})^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Recurrence and bounds

1. [0.5 pt] v $_1 = \sqrt{72 + 0} = \sqrt{72}$ and $v_2 = \sqrt{72 + \sqrt{72}}$.

2. [1.5 pt] Note: $P(n)$: « v_n is defined and $0 \leq v_n \leq 9$ ».

   Initialization. $v_0 = 0$ is defined and belongs to $[0, 9]$; $P(0)$ is true.

   Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True, then.

$$0 \leq 72 \leq 72 + v_n \leq 72 + 9 = 81.$$

The square root is therefore defined and, by growth of this function on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{72 + v_n} \leq \sqrt{81} = 9.$$

So $P(n+1)$ is true.

Conclusion. By induction, all terms are defined and belong to $[0, 9]$.

3. [1.5 pt] For $n \in \mathbb{N}$, posisons $R(n)$: « $v_{n+1} \geq v_n$ ».

Initialization. $v_1 = \sqrt{72} > 0 = v_0$; $R(0)$ is true.

Heredity. Fixes $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 72 and then applying growth square root,

$$v_{n+2} = \sqrt{72 + v_{n+1}} \geq \sqrt{72 + v_n} = v_{n+1}.$$

It's $R(n+1)$.

Conclusion. By induction, $v_{n+1} \geq v_n$ for every $n \geq 0$: (v_n) is increasing.

4. [1 pt] Note: $S(n)$: « $v_n < 9$". In row 0, $v_0 = 0 < 9$. Fixes $n \geq 0$ and suppose $S(n)$ True. Since radicandes are positive, the strict growth of the square root gives

$$v_{n+1} = \sqrt{72 + v_n} < \sqrt{72 + 9} = \sqrt{81} = 9.$$

So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n < 9$ for all $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false in 0.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 4 $\vec{u} + 5\vec{v} = 4(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 5\vec{\jmath} + 9\vec{k} = \vec{w}$.

3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 4, \qquad \mu = 5, \qquad \lambda + \mu = 10.$$

The first two equalitys give $\lambda + \mu = 9$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $5\vec{u} - 4\vec{v} = 5(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 4\vec{\jmath} + \vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{11} = \frac{\overrightarrow{BD}}{11}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two vectors of the base are not colinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of decomposition in this base.

4. [1 pt] The diagonals of a parallelepipeda cut in the midpoint. $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$ which is proves $O \in (AG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{11}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{11}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{11} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS1 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 2$ and, for the integer natural n, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $2 \leq u_n < 9$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 9$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 10$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+7\vec{\jmath}+10\vec{k}, \quad \vec{z}=3\vec{\imath}+7\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 7 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS1 · Zero 2

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{36}{5} = \frac{38}{5}, \qquad u_2 = \frac{1}{5} \times \frac{38}{5} + \frac{36}{5} = \frac{218}{25}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 9 - 7\left(\frac{1}{5}\right)^n$.

Initialization. For $n = 0$, $9 - 7 \times 1 = 2 = u_0$. So $P(0)$ is true.

Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{36}{5} \\ &= \frac{1}{5}\left[9 - 7\left(\frac{1}{5}\right)^n\right] + \frac{36}{5} \\ &= \frac{9}{5} + \frac{36}{5} - 7\left(\frac{1}{5}\right)^{n+1} \\ &= 9 - 7\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. The principle of recurrence gives $u_n = 9 - 7\left(\frac{1}{5}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{5} < 1$, for everything $n \in \mathbb{N}$ We have $0. < (\frac{1}{5})^n \leq 1$. Multiplication by $-7 < 0$ reverse inequalities:

$$-7 \leq -7(\frac{1}{5})^n < 0.$$

By adding 9 and using the previous formula, you get $2 \leq u_n < 9$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{5})^{n+1} + 7(\frac{1}{5})^n \\ &= 7(1 - \frac{1}{5})(\frac{1}{5})^n = \frac{28}{5}(\frac{1}{5})^n > 0. \end{aligned}$$

The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $9 - u_n = 7(\frac{1}{5})^n > 0$. No term is therefore equal to 9, regardless of the number of steps.

Exercise 2 · Recurrence and bounds

1. [0.5 pt] v $_1 = \sqrt{90 + 0} = \sqrt{90}$ and $v_2 = \sqrt{90 + \sqrt{90}}$.

2. [1.5 pt] Note: $P(n)$: « v_n is defined and $0 \leq v_n \leq 10$ ».

Initialization. $v_0 = 0$ is defined and belongs to $[0, 10]$; $P(0)$ is true.

Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True, then.

$$0 \leq 90 \leq 90 + v_n \leq 90 + 10 = 100.$$

The square root is therefore defined and, by growth of this function on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{90 + v_n} \leq \sqrt{100} = 10.$$

So $P(n+1)$ is true.

Conclusion. By induction, all terms are defined and belong to $[0, 10]$.

3. [1.5 pt] For $n \in \mathbb{N}$, posisons $R(n)$: « $v_{n+1} \geq v_n$ ».

Initialization. $v_1 = \sqrt{90} > 0 = v_0$; $R(0)$ is true.

Heredity. Fixes $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 90 and then applying growth square root,

$$v_{n+2} = \sqrt{90 + v_{n+1}} \geq \sqrt{90 + v_n} = v_{n+1}.$$

It's $R(n+1)$.

Conclusion. By induction, $v_{n+1} \geq v_n$ for every $n \geq 0$: (v_n) is increasing.

4. [1 pt] Note: $S(n)$: « $v_n < 10$". In row 0, $v_0 = 0 < 10$. Fixes $n \geq 0$ and suppose $S(n)$ True. Since the radicandes are positive, the strict growth of the square root gives

$$v_{n+1} = \sqrt{90 + v_n} < \sqrt{90 + 10} = \sqrt{100} = 10.$$

So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n < 10$ for all $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false in 0.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 3 $\vec{u} + 7\vec{v} = 3(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 7\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 3, \qquad \mu = 7, \qquad \lambda + \mu = 11.$$

The first two equalitys give $\lambda + \mu = 10$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $7\vec{u} - 3\vec{v} = 7(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 3\vec{\jmath} + 4\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two vectors of the base are not colinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of decomposition in this base.

4. [1 pt] The diagonals of a parallelepipeda cut in the midpoint. $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$ which is proves $O \in (AG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS1 · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A law to be found — 4 points

We model a grandeur by $u_{n+1} = au_n + b$, for $n \in \mathbb{N}$. We observe $u_0 = 2$, $u_1 = 5$ and $u_2 = \frac{13}{2}$.

1. Determine a and b. (1 pt)

2. Demonstrate by induction that $u_n = 8 - 6\left(\frac{1}{2}\right)^n$. 2 pts

3. Deduce the direction of variation and a upper bound from the sequence. (1 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 11$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · A · continued

Exercise 3 · A coefficient that changes coplanarity — 6 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$, we're laying $u=\vec{\imath}+\vec{\jmath}$, $v=\vec{\jmath}+\vec{k}$ and $w_m=\vec{\imath}+m\vec{\jmath}+2\vec{k}$ where: $m\in\mathbb{R}$.

1. Justify that u, v are not colinear. (1 pt)

2. Determine the values of m for which w_m is a combination of u, v. 2 pts

3. One student claims that two coefficients equals are always sufficient. Explain the error by taking $m=0$. 2 pts

4. Give the combination obtained for the eligible value of m. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath},\qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath},\qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS1 · A

Exercise 1 · A law to be found

1. $2a+b=5$ and $5a+b=13/2$; the difference gives $a=1/2$, then $b=4$.

2. Note: $P(n)$: $u_n=8-6(1/2)^n$. Initialization: $8-6=2=u_0$. Fixing $n\geq 0$ and suppose $P(n)$. Then $u_{n+1}=\frac{1}{2}[8-6(1/2)^n]+4=8-6(1/2)^{n+1}$. It's $P(n+1)$. In induction, the formula is for every $n\in\mathbb{N}$. Scale: initialization and ownership 0.5; inheritance 1; conclusion 0.5.

3. $u_{n+1}-u_n=3(1/2)^n>0$; $u_n<8$ because $6(1/2)^n>0$. Two justifications at 0.5 points.

Exercise 2 · Recurrence and bounds

1. [0.5 pt] v $_1=\sqrt{110+0}=\sqrt{110}$ and $v_2=\sqrt{110+\sqrt{110}}$.

2. [1.5 pt] Note: $P(n)$: « v_n is defined and $0\leq v_n\leq 11$ ».

   Initialization. $v_0=0$ is defined and belongs to $[0,11]$; $P(0)$ is true.

   Heredity. Fixes $n\in\mathbb{N}$ and suppose $P(n)$ True, then.

   $$0\leq 110\leq 110+v_n\leq 110+11=121.$$

   The square root is therefore defined and, by growth of this function on $[0,+\infty[$,

   $$0\leq v_{n+1}=\sqrt{110+v_n}\leq\sqrt{121}=11.$$

   So $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,11]$.

3. [1.5 pt] For $n\in\mathbb{N}$, posisons $R(n)$: « $v_{n+1}\geq v_n$ ».

   Initialization. $v_1=\sqrt{110}>0=v_0$; $R(0)$ is true.

   Heredity. Fixes $n\geq 0$ and suppose $v_{n+1}\geq v_n$. Adding 110 and then applying growth square root,

   $$v_{n+2}=\sqrt{110+v_{n+1}}\geq\sqrt{110+v_n}=v_{n+1}.$$

   It's $R(n+1)$.

   Conclusion. By induction, $v_{n+1}\geq v_n$ for every $n\geq 0$: (v_n) is increasing.

4. [1 pt] Note: $S(n)$: « $v_n<11$". In row 0, $v_0=0<11$. Fixes $n\geq 0$ and suppose $S(n)$ True. Since the radicandes are positive, the strict growth of the square root gives

   $$v_{n+1}=\sqrt{110+v_n}<\sqrt{110+11}=\sqrt{121}=11.$$

   So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n<11$ for all $n\in\mathbb{N}$.

5. [0.5 pt] No. For example $P(n)$: $n\geq 1$ is true for all integers $n\geq 1$ but false in 0.

Exercise 3 · A coefficient that changes coplanarity

1. In $v=tu$, the coefficient of $\vec{k}$ would give $1=0$.

2. $w_m=\alpha u+\beta v$ imposes $\alpha=1$, $\beta=2$ and $m=\alpha+\beta=3$. This value satisfies the three coefficients. System 1 point, resolution and verification 1 point.

3. For $m = 0$, the coefficients $\vec{\imath}$ and $\vec{k}$ and 2, but the coefficient of $\vec{\jmath}$ Would be 3 instead of 0. A contradiction in a single equation invalidates the combination. Calculation 1 point, explanation 1 point.

4. $w_3 = u + 2v$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two vectors of the base are not colinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of decomposition in this base.

4. [1 pt] The diagonals of a parallelepipeda cut in the midpoint. $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$which is proves $O \in (AG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS1 · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Repair an induction — 4 points

We define $u_0 = 1$ and $u_{n+1} = 3u_n + 2$ for $n \in \mathbb{N}$. A student proposes: " $u_n = 3^n$. That's true in row 0. If that's true in the row n, then $u_{n+1} = 3^{n+1} + 2$, so it's true in the next row."

1. Identify precisely the error and give a term that refutes the formula. (1 pt)

2. Demonstrate by induction that $u_n = 2 \times 3^n - 1$. 2 pts

3. Determine the first ranking for which $u_n > 100$. (1 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 7$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · B · continued

Exercise 3 · Describe a vector construction — 6 points

We're giving a base $(\vec{\imath},\vec{\jmath},\vec{k})$. A tetrahedron $ABCD$ is defined by $\overrightarrow{AB}=\vec{\imath}$, $\overrightarrow{AC}=\vec{\jmath}$, $\overrightarrow{AD}=\vec{k}$. We place I, J by $\overrightarrow{AI}=2\vec{\imath}+\vec{\jmath}$ and $\overrightarrow{AJ}=\vec{\jmath}+\vec{k}$.

1. Calculate $\overrightarrow{IJ}$. (1 pt)

2. Vectors $\overrightarrow{AI},\overrightarrow{AJ},\overrightarrow{AD}$ Are they coplanaries? 2 pts

3. Definition K by $\overrightarrow{AK}=2\overrightarrow{AJ}-\overrightarrow{AI}$. Show that I, J, K are aligned. 2 pts

4. Please specify the point in the midpoint of the other two. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{15}\vec{\imath},\qquad \overrightarrow{AJ}=\frac{1}{15}\vec{\jmath},\qquad \overrightarrow{AK}=\frac{1}{15}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS1 · B

Exercise 1 · Repair an induction

1. The result of inheritance is not the property to rank $n+1$: the term $+$Two left. $u_1 = 5 \neq 3$.

2. Poisons $P(n)$: $u_n = 2 \times 3^n - 1$. In row 0, $2 \quad -1 = 1$. Suppose $P(n)$ at one rank $n \geq 0$ fixed; $u_{n+1} = 3(2 \times 3^n - 1) + 2 = 2 \times 3^{n+1} - 1$. By induction, $P(n)$ is valid at any level. Scale: 0.5; 1; 0.5 for all three stages.

3. $u_3 = 53 \leq 100 < u_4 = 161$. The formula shows that u_n is strictly increasing, so the first row is 4.

Exercise 2 · Recurrence and bounds

1. [0.5 pt] v $_1 = \sqrt{42+0} = \sqrt{42}$ and $v_2 = \sqrt{42+\sqrt{42}}$.

2. [1.5 pt] Note: $P(n)$: « v_n is defined and $0 \leq v_n \leq 7$ ».

   Initialization. $v_0 = 0$ is defined and belongs to $[0,7]$; $P(0)$ is true.

   Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True, then.

   $$0 \leq 42 \leq 42 + v_n \leq 42 + 7 = 49.$$

   The square root is therefore defined and, by growth of this function on $[0, +\infty[$,

   $$0 \leq v_{n+1} = \sqrt{42+v_n} \leq \sqrt{49} = 7.$$

   So $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,7]$.

3. [1.5 pt] For $n \in \mathbb{N}$, posisons $R(n)$: « $v_{n+1} \geq v_n$ ».

   Initialization. $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   Heredity. Fixes $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 42 and then applying growth square root,

   $$v_{n+2} = \sqrt{42+v_{n+1}} \geq \sqrt{42+v_n} = v_{n+1}.$$

   It's $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \geq v_n$ for every $n \geq 0$: (v_n) is increasing.

4. [1 pt] Note: $S(n)$: « $v_n < 7$". In row 0, $v_0 = 0 < 7$. Fixes $n \geq 0$ and suppose $S(n)$ True. Since radicandes are positive, the strict growth of the square root gives

   $$v_{n+1} = \sqrt{42+v_n} < \sqrt{42+7} = \sqrt{49} = 7.$$

   So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n < 7$ for all $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false in 0.

Exercise 3 · Describe a vector construction

1. By Chasles, $\overrightarrow{IJ} = -2\vec{\imath} + \vec{k}$.

2. AI, AJ are not colinear (coefficients of $\vec{\imath}$ and $\vec{k}$) In $\vec{k} = \alpha(2\vec{\imath} + \vec{\jmath}) + \beta(\vec{\jmath} + \vec{k})$, coefficients require $\alpha = 0$, $\beta = 1$ and $\alpha + \beta = 0$, contradiction. Non-coplanarity. 1 point for the system, 1 for the contradiction.

3. $\overrightarrow{IK} = \overrightarrow{AK} - \overrightarrow{AI} = 2(\overrightarrow{AJ} - \overrightarrow{AI}) = 2\overrightarrow{IJ}$. This vector is non-zero; the three points are aligned. 1 point, justified conclusion 1 point.

4. J is the midpoint of $[IK]$ since $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{IK}$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{15} = \frac{\overrightarrow{BD}}{15}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two vectors of the base are not colinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of decomposition in this base.

4. [1 pt] The diagonals of a parallelepipeda cut in the midpoint. $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$which is proves $O \in (AG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{15}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{15}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{15} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS2 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ and $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. (1 pt)

2. Follow-up $u_n = 2 + (-1)^n$ converging to 2. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+6\vec{\jmath}+8\vec{k}, \quad \vec{z}=2\vec{\imath}+6\vec{\jmath}+9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 6 $\vec{u}-2\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the rights (AB) and (CG) : are they parallel, stinging or not Coplanaries? Justify. (1 pt)

4. Note O the center of the parallelepipeda. O Does it belong to the plan? (ABG) ? Justify. (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS2 · Zero 1

Exercise 1 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{4}{n} - \frac{1}{n^2}}{4 + \frac{3}{n^2}}.$$

Terms $\frac{1}{n}$ and $\frac{1}{n^2}$ The numerator tends to 3 and the denominator to 4 $\neq 0$; by quotient, $u_n \to \frac{3}{4}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2 + 4n) - n^2}{\sqrt{n^2 + 4n} + n} = \frac{4n}{n(\sqrt{1 + \frac{4}{n}} + 1)} = \frac{4}{\sqrt{1 + \frac{4}{n}} + 1}.$$

3. [1 pt] The denominator tends to 2; limit 2.

4. [1 pt] w $_n = n^2(3 - \frac{4}{n}) \to +\infty$; the second factor tends to 3 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 2 · True or false: reasoning on limits

1. [1 pt] True: the denominator has a non-zero limit; the quotient limit is $-\frac{2}{5}$.

2. [1 pt] False: the same terms are worth 3, the odd terms 1. The two values remain at a distance 1 of 2.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1 + \frac{3}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, $2 \quad \vec{u} + 6\vec{v} = 2(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 2\vec{\imath} + 6\vec{\jmath} + 8\vec{k} = \vec{w}$.

3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 2, \qquad \mu = 6, \qquad \lambda + \mu = 9.$$

The first two equalitys give $\lambda + \mu = 8$, in contradiction to the third. vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 2\vec{v} = 6(\vec{\imath} + \vec{k}) - 2(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 2\vec{\jmath} + 4\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] Directors $\vec{\imath}$ and $\vec{k}$, non-colonial. An intersection would impose $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. [1 pt] $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, therefore $O \in (AG) \subset (ABG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS2 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ and $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. (1 pt)

2. Follow-up $u_n = 5 + (-1)^n$ converging to 5. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. (1 pt)

DS2 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+7\vec{\jmath}+11\vec{k}, \quad \vec{z}=4\vec{\imath}+7\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 7 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the rights (AB) and (CG) : are they parallel, stinging or not Coplanaries? Justify. (1 pt)

4. Note O the center of the parallelepipeda. O Does it belong to the plan? (ABG) ? Justify. (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Corrected · DS2 · Zero 2

Exercise 1 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{6}{n} - \frac{1}{n^2}}{3 + \frac{1}{n^2}}.$$

Terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to 3 $\neq 0$; by quotient, $u_n \to \frac{2}{3}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2 + 6n) - n^2}{\sqrt{n^2 + 6n} + n} = \frac{6n}{n(\sqrt{1 + \frac{6}{n}} + 1)} = \frac{6}{\sqrt{1 + \frac{6}{n}} + 1}.$$

3. [1 pt] The denominator tends to 2; limit 3.

4. [1 pt] w $_n = n^2(2 - \frac{6}{n}) \to +\infty$; the second factor tends to 2 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 2 · True or false: reasoning on limits

1. [1 pt] True: the denominator has a non-zero limit; the quotient limit is $-\frac{5}{6}$.

2. [1 pt] False: the same terms are equal to 6, the odd terms 4. The two values remain at distance 1 of 5.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1 + \frac{1}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{6}{n}$; $n > 600$ gives $\frac{6}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 4 $\vec{u} + 7\vec{v} = 4(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 7\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 4, \qquad \mu = 7, \qquad \lambda + \mu = 12.$$

The first two equalitys give $\lambda + \mu = 11$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $7\vec{u} - 4\vec{v} = 7(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 4\vec{\jmath} + 3\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepipeda

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are non-zero and colinear: (IJ) and (BD) are parallel.

3. [1 pt] Directors $\vec{\imath}$ and $\vec{k}$, non-colonial. An intersection would impose $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. [1 pt] $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, therefore $O \in (AG) \subset (ABG)$.

5. [1 pt] Points check $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, The vectors are not collinear, so I, J, K are not aligned.

DS2 · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Accuracy expressed by a band — 4 points

For $n \in \mathbb{N}$, we're laying $u_n = 4 + \frac{6}{n+2}$.

1. Calculate the limit and justify why u_n Never worth that limit. (1 pt)

2. Find the first row from which $|u_n - 4| < 0.01$. Check the bound. 2 pts

3. Correctly deny the phrase "all terms are strictly less than 5", then give a Witness. (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 4$ and $v_n \to -8$, then $\frac{u_n}{v_n} \to -\frac{1}{2}$. (1 pt)

2. Follow-up $u_n = 4 + (-1)^n$ converging to 4. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+2n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{4n+8}{n}$, we have $|u_n - 4| < 0.01$ as soon as $n > 800$. (1 pt)

DS2 · A · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+4\vec{\jmath}+7\vec{k}, \quad \vec{z}=3\vec{\imath}+4\vec{\jmath}+8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 4 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Visible or demonstrated intersection — 6 points

In a space coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$, $D(0;0;4)$. We're laying $I(2;0;0)$, $J(0;2;0)$, $K(0;0;2)$ and $G(1;1;1)$.

1. Show that I, J, K are not aligned. (1 pt)

2. Determine whether G belongs to the plan (IJK) a linear combination. 2 pts

3. Righteousness (IJ) and (BC) Are they strictly parallel? 2 pts

4. Explain why the projection of G inside the drawing triangle would not be sufficient to prove his membership in the plan. (1 pt)

Fixed · DS2 · A

Exercise 1 · Accuracy expressed by a band

1. $6/(n+2) \to 0$, therefore $u_n \to 4$; the quotient is always strictly positive.
2. $6/(n+2) < 1/100$ is equal to $n+2 > 600$, or $n \geq 599$. In row 598, the difference is 0,01 ; at rank 599 it is strictly smaller, and then it decreases. Resolution 1 point, minimality 1 point.
3. "There is a rank n such that $u_n \geq 5$". For example $u_0 = 7$.

Exercise 2 · True or false: reasoning on limits

1. [1 pt] True: the denominator has a non-zero limit; the quotient limit is $-\frac{1}{2}$.
2. [1 pt] False: the same terms are equal to 5, the odd terms 3. The two values remain at distance 1 of 4.
3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.
4. [1 pt] True: $w_n = \sqrt{1+\frac{2}{n}} \to 1$.
5. [1 pt] True: the difference is $\frac{8}{n}$; $n > 800$ gives $\frac{8}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.
2. [0.5 pt] By developing, 3 $\vec{u} + 4\vec{v} = 3(\vec{\imath}+\vec{k}) + 4(\vec{\jmath}+\vec{k}) = 3\vec{\imath}+4\vec{\jmath}+7\vec{k} = \vec{w}$.
3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 3, \qquad \mu = 4, \qquad \lambda + \mu = 8.$$

   The first two equalitys give $\lambda + \mu = 7$, in contradiction with the third. vectors are therefore not coplanar.
4. [1 pt] $4\vec{u} - 3\vec{v} = 4(\vec{\imath}+\vec{k}) - 3(\vec{\jmath}+\vec{k}) = 4\vec{\imath} - 3\vec{\jmath} + \vec{k}$.
5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Visible or demonstrated intersection

1. $IJ = (-2;2;0)$ and $IK = (-2;0;2)$ are not colinear, by their second and third coordinates.
2. $IG = (-1;1;1)$. Equality $IG = sIJ + tIK$ imposes $s = t = 1/2$ but the first coordinated would be $-2 \neq -1$. Therefore $G \notin (IJK)$. 1 system point, 1 contradiction.
3. $BC = (-4;4;0) = 2IJ$. In addition $BI = (-2;0;0)$ is not colinear to BC : $I \notin (BC)$. Both straight are distinct and parallel. 1 point for each condition.
4. The projection loses depth; several points of space can have the same image on the leaf.

DS2 · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · When two sequences are almost no longer distinguishable — 4 points

For $n \geq 1$, $a_n = \frac{3n+1}{n+2}$ and $b_n = \frac{3n-4}{n+2}$.

1. Calculate their limits. (1 pt)

2. Resolve $|a_n - b_n| < 0.05$ and rank first. 2 pts

3. Refuse "two sequences that have the same limit have equal terms from a certain rank." (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 3$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{3}{5}$. (1 pt)

2. Follow-up $u_n = 3 + (-1)^n$ converging to 3. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{3n+5}{n}$, we have $|u_n - 3| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · B · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+6\vec{\jmath}+9\vec{k}, \quad \vec{z}=3\vec{\imath}+6\vec{\jmath}+10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 6 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Two descriptions of the same plan — 6 points

In a coordinate system, $A(1;0;0)$, $B(1;1;0)$, $C(1;0;1)$, $D(1;2;-1)$, $M(2;2;-1)$.

1. Show that A, B, C define a plan. (1 pt)

2. Show that D belongs to this plan and determines the coefficients of the combination used. (2 pts)

3. Can one replace C by D Without changing the plan? 2 pts

4. Item M Does it belong to the plan? (1 pt)

Corrected · DS2 · B

Exercise 1 · When two sequences are almost no longer distinguishable

1. By dividing by n, each tends to 3 (0.5 points each).

2. $a_n - b_n = 5/(n+2) > 0$. Inequality is equal to $n+2 > 100$, so $n \geq 99$. In row 98, difference is exactly 0.05. Calculation of the deviation 1 point, resolution and bound 1 point.

3. These two sequences have the same limit but their difference 5 $/(n+2)$ never cancels: they This is a complete counter-example.

Exercise 2 · True or false: reasoning on limits

1. [1 pt] True: the denominator has a non-zero limit; the quotient limit is $-\frac{3}{5}$.

2. [1 pt] False: the same terms are worth 4, the odd terms 2. The two values remain at a distance 1 of 3.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1 + \frac{3}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a base

1. [1 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 3 $\vec{u} + 6\vec{v} = 3(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 6\vec{\jmath} + 9\vec{k} = \vec{w}$.

3. [1.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 3, \qquad \mu = 6, \qquad \lambda + \mu = 10.$$

The first two equalitys give $\lambda + \mu = 9$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 3\vec{v} = 6(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 3\vec{\jmath} + 3\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Two descriptions of the same plan

1. $AB = (0;1;0)$ and $AC = (0;0;1)$ are not colinear.

2. $AD = (0;2;-1) = 2AB - AC$. The combination proves $D \in (ABC)$. Coefficients 1 point, justification 1 point.

3. AB, AD are not colinear since AD has a third non-zero coordinate. $AC = 2AB - AD$. The two pairs generate the same plane by A. 1 point for each argument.

4. $AM = (1;2;-1)$ has a first non-zero coordinate, while any combination of AB, AC No, no, no, no.

Common duty · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 2$ and, for the integer natural n, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $2 \leq u_n < 8$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 8$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · Zero 1 · continued

Exercise 3 · Limitations and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{2x+6}{x-2}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 2^- and 2^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ justifying. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet that contains 15 symbols
The rules on repetition are specified in each question, independently of the
other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)

2. How many codes can be formed if each symbol is to be used at most once? (1 pt)

3. A symbol of the alphabet is fixed in advance. Repetitions are allowed, how many codes contain this symbol at least once? (1 pt)

4. The first symbol of the code is imposed and repetitions are prohibited.
Can we train? (1 pt)

5. Two distinct symbols of the alphabet are prohibited. Other symbols can be repeated.
How many codes can we train? (1 pt)

Corrected · Common duty · Zero 1

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{4} \times 2 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{4} \times \frac{13}{2} + 6 = \frac{61}{8}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 8 - 6\left(\frac{1}{4}\right)^n$.

Initialization. For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. So $P(0)$ is true.

Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{4}u_n + 6 \\ &= \frac{1}{4}\left[8 - 6\left(\frac{1}{4}\right)^n\right] + 6 \\ &= 2 + 6 - 6\left(\frac{1}{4}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{4}\right)^{n+1}. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. The principle of recurrence gives $u_n = 8 - 6\left(\frac{1}{4}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{4} < 1$, for everything $n \in \mathbb{N}$ We have $0. < (\frac{1}{4})^n \leq 1$. Multiplication by $-6 < 0$ reverse inequalities:

$$-6 \leq -6(\frac{1}{4})^n < 0.$$

By adding 8 and using the previous formula, you get $2 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6(\frac{1}{4})^{n+1} + 6(\frac{1}{4})^n \\ &= 6(1 - \frac{1}{4})(\frac{1}{4})^n = \frac{9}{2}(\frac{1}{4})^n > 0. \end{aligned}$$

The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 6(\frac{1}{4})^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{5}{n} - \frac{1}{n^2}}{4 + \frac{2}{n^2}}.$$

Terms $\frac{1}{n}$ and $\frac{1}{n^2}$ The numerator tends to 3 and the denominator to 4 $\neq 0$; by quotient, $u_n \to \frac{3}{4}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. [1 pt] The denominator tends to 2 ; limit $\frac{5}{2}$.

4. [1 pt] w $_n = n^2(3-\frac{5}{n}) \to +\infty$; the second factor tends to 3 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 3 · Limitations and asymptotes

1. [1 pt] The denominator cancels only in 2: $D_f = \mathbb{R} \setminus \{2\}$. Like 2 $x+6 = 2(x-2)+10$, we get $f(x) = 2 + \frac{10}{x-2}$.

2. [1 pt] The two limits are 2.

3. [1 pt] When $x \to 2^-$, $x-2 \to 0$ by negative values; as $10 > 0$, $\frac{10}{x-2} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 2$ and $y = 2$; curve below for $x < 2$ and above for $x > 2$.

5. [1 pt] Poisons $t = 2x$. Then $t \to +\infty$ and $x^2e^{-2x} = \frac{1}{4}\frac{t^2}{e^t}$. Comparative growth gives $\frac{t^2}{e^t} \to 0$; the limit sought therefore is 0.

Exercise 4 · Codes and selections

1. [1 pt] $15^4 = 50625$ by the multiplicative principle.

2. [1 pt] Successive choices have 15, 14, then a choice of fewer at each stage, up to 12 possibilities. By multiplication,

$$15 \times 14 \times 13 \times 12 = 32760.$$

The order of symbols is part of the code.

3. [1 pt] $15^4 - 14^4 = 12209$ per supplement.

4. [1 pt] The symbol of the first position is fixed and can no longer be used. the following give successively 14, . . . , 12 choices:

$$14 \times 13 \times 12 = 2184.$$

5. [1 pt] $13^4 = 28561$.

Common duty · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 3$ and, for the integer natural n, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $3 \leq u_n < 10$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 10$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ and $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{7}{\sqrt{1+\frac{7}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · Zero 2 · continued

Exercise 3 · Limitations and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{5x+7}{x-3}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 3^- and 3^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ justifying. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet that contains 16 symbols
The rules on repetition are specified in each question, independently of the
other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)

2. How many codes can be formed if each symbol is to be used at most once? (1 pt)

3. A symbol of the alphabet is fixed in advance. Repetitions are allowed, how many codes contain this symbol at least once? (1 pt)

4. The first symbol of the code is imposed and repetitions are prohibited.
Can we train? (1 pt)

5. Two distinct symbols of the alphabet are prohibited. Other symbols can be repeated.
How many codes can we train? (1 pt)

Corrected · Common duty · Zero 2

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{3} \times 3 + \frac{20}{3} = \frac{23}{3}, \qquad u_2 = \frac{1}{3} \times \frac{23}{3} + \frac{20}{3} = \frac{83}{9}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 10 - 7\left(\frac{1}{3}\right)^n$.

Initialization. For $n = 0$, $10 - 7 \times 1 = 3 = u_0$. So $P(0)$ is true.

Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{3}u_n + \frac{20}{3} \\ &= \frac{1}{3}\left[10 - 7\left(\frac{1}{3}\right)^n\right] + \frac{20}{3} \\ &= \frac{10}{3} + \frac{20}{3} - 7\left(\frac{1}{3}\right)^{n+1} \\ &= 10 - 7\left(\frac{1}{3}\right)^{n+1}. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. The principle of recurrence gives $u_n = 10 - 7\left(\frac{1}{3}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{3} < 1$, for everything $n \in \mathbb{N}$ We have $0. < \left(\frac{1}{3}\right)^n \leq 1$. Multiplication by $-7 < 0$ reverse inequalities:

$$-7 \leq -7\left(\frac{1}{3}\right)^n < 0.$$

By adding 10 and using the previous formula, you get $3 \leq u_n < 10$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7\left(\frac{1}{3}\right)^{n+1} + 7\left(\frac{1}{3}\right)^n \\ &= 7\left(1 - \frac{1}{3}\right)\left(\frac{1}{3}\right)^n = \frac{14}{3}\left(\frac{1}{3}\right)^n > 0. \end{aligned}$$

The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $10 - u_n = 7\left(\frac{1}{3}\right)^n > 0$. No term is So equal to 10, regardless of the number of steps.

Exercise 2 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{7}{n} - \frac{1}{n^2}}{3 + \frac{3}{n^2}}.$$

Terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to $3 \neq 0$; by quotient, $u_n \to \frac{2}{3}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2+7n)-n^2}{\sqrt{n^2+7n}+n} = \frac{7n}{n(\sqrt{1+\frac{7}{n}}+1)} = \frac{7}{\sqrt{1+\frac{7}{n}}+1}.$$

3. [1 pt] The denominator tends to 2 ; limit $\frac{7}{2}$.

4. [1 pt] w $_n = n^2(2-\frac{7}{n}) \to +\infty$; the second factor tends to 2 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 3 · Limitations and asymptotes

1. [1 pt] The denominator cancels only in 3 : $D_f = \mathbb{R} \setminus \{3\}$. Like 5 $x+7 = 5(x-3)+22$, we get $f(x) = 5 + \dfrac{22}{x-3}$.

2. [1 pt] The two limits are worth 5.

3. [1 pt] When $x \to 3^-$, $x-3 \to 0$ by negative values; like 22 > 0, $\frac{22}{x-3} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 3$ and $y = 5$; curve below for $x < 3$ and above for $x > 3$.

5. [1 pt] Poisons $t = 3x$. Then $t \to +\infty$ and $x^5 e^{-3x} = \frac{1}{243}\frac{t^5}{e^t}$. Comparative growth gives $\frac{t^5}{e^t} \to 0$; the limit sought therefore is 0.

Exercise 4 · Codes and selections

1. [1 pt] $16^5 = 1048576$ by the multiplicative principle.

2. [1 pt] Successive choices have 16, 15, then a choice of minus at each stage, up to 12 possibilities. By multiplication,

$$16 \times 15 \times 14 \times 13 \times 12 = 524160.$$

The order of symbols is part of the code.

3. [1 pt] $16^5 - 15^5 = 289201$ per supplement.

4. [1 pt] The symbol of the first position is fixed and can no longer be used. the following give successively 15, . . . , 12 choices:

$$15 \times 14 \times 13 \times 12 = 32760.$$

5. [1 pt] $14^5 = 537824$.

Common duty · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 1$ and, for the integer natural n, $u_{n+1} = \frac{1}{2}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 12 - 11\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $1 \leq u_n < 12$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 12$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{5n^2+8n-1}{6n^2+1}$ and $v_n = \sqrt{n^2+8n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{8}{\sqrt{1+\frac{8}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 5n^2 - 8n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · A · continued

Exercise 3 · Identify a model from its asymptotes — 4 points

We're looking for f in the form $f(x) = a + \frac{b}{x-1}$, $b \neq 0$. Its curve admits horizontal asymptote $y = 2$ and passes through $(0; -1)$.

1. Determine a, b and the domain. (1 pt)

2. Calculate the limits in 1^- and 1^+. (1 pt)

3. Study the position of the curve in relation to $y = 2$ and resolve $f(x) = 5$. 2 pts

Exercise 4 · Codes whose rules change — 6 points

A code contains four digits taken from 0, 1, 2, 3, 4. Zero may be in the first position.

1. Count all codes with repetition. (1 pt)

2. Count those without repetition. (1 pt)

3. Count those that contain exactly a zero, the other figures can be repeated. (2 pts)

4. Count those that contain at least one zero by two methods or an explained method in detail. 2 pts

Corrected · Common duty · A

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{2} \times \frac{13}{2} + 6 = \frac{37}{4}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 12 - 11\left(\frac{1}{2}\right)^n$.

Initialization. For $n = 0$, $12 - 11 \times 1 = 1 = u_0$. So $P(0)$ is true.

Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 6 \\ &= \frac{1}{2}\left[12 - 11\left(\frac{1}{2}\right)^n\right] + 6 \\ &= 6 + 6 - 11\left(\frac{1}{2}\right)^{n+1} \\ &= 12 - 11\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. The principle of recurrence gives $u_n = 12 - 11\left(\frac{1}{2}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{2} < 1$, for everything $n \in \mathbb{N}$ We have $0. < (\frac{1}{2})^n \leq 1$. Multiplication by $-11 < 0$ reverse inequalities:

$$-11 \leq -11(\frac{1}{2})^n < 0.$$

By adding 12 and using the previous formula, you get $1 \leq u_n < 12$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -11(\frac{1}{2})^{n+1} + 11(\frac{1}{2})^n \\ &= 11(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{11}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $12 - u_n = 11(\frac{1}{2})^n > 0$. No term is So equal to 12, regardless of the number of steps.

Exercise 2 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{5 + \frac{8}{n} - \frac{1}{n^2}}{6 + \frac{1}{n^2}}.$$

Terms $\frac{1}{n}$ and$\frac{1}{n^2}$ The numerator tends to 5 and the denominator to 6 $\neq 0$; by quotient, $u_n \to \frac{5}{6}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2+8n)-n^2}{\sqrt{n^2+8n}+n} = \frac{8n}{n(\sqrt{1+\frac{8}{n}}+1)} = \frac{8}{\sqrt{1+\frac{8}{n}}+1}.$$

3. [1 pt] The denominator tends to 2; limit 4.

4. [1 pt] w $_n = n^2(5-\frac{8}{n}) \to +\infty$; the second factor tends to 5 > 0.

5. [1 pt] No: $(2n)-n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 3 · Identify a model from its asymptotes

1. The limit to infinity is a, therefore $a = 2$. Then $2-b=-1$ given $b=3$. Area: $\mathbb{R}\setminus\{1\}$.

2. The quotient $3/(x-1)$ tend towards $-\infty$ left and $+\infty$ on the right (0.5 points per side).

3. $f-2 = 3/(x-1)$: below for $x<1$, above for $x>1$, never on asymptote (1 pt).
$f=5 \iff 3/(x-1)=3 \iff x=2$, eligible (1 pt).

Exercise 4 · Codes whose rules change

1. $5^4 = 625$.

2. $5\times4\times3\times2 = 120$.

3. Four positions for zero, then four choices for each of the other three positions: 4 $\times 4^3 = 256$.
Position 1 point, filling 1 point.

4. For additional information: $5^4 - 4^4 = 369$; the codes using only 1, 2, 3, 4 are removed or according to the number of zeros: 256 $+ 6\times16 + 4\times4 + 1 = 369$. The coefficient 6 is located by listing the six pairs
Model 1 point, 1 point calculation.

Common duty · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 2$ and, for the integer natural n, $u_{n+1} = \frac{1}{5}u_n + \frac{32}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $2 \leq u_n < 8$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 8$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{4n^2+5n-1}{5n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 4n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · B · continued

Exercise 3 · One root and two infinites — 4 points

It is considered $g(x) = \sqrt{x^2+4} - x$ on $\mathbb{R}$.

1. Show that $g(x) > 0$ on $\mathbb{R}$. (1 pt)

2. Determine the limit in $+\infty$ by a justified transformation. 2 pts

3. Determining the limit in $-\infty$ and explain why the previous result does not translate. (1 pt)

Exercise 4 · Build a decision tree — 6 points

An outfit combines a shirt among 3, pants among 4 and shoes among 2.
shirts is red, one of the pants is red. The items are all distinct.

1. Count outfits without restriction. (1 pt)

2. Count outfits with red shirt and red pants. (1 pt)

3. Count the outfits with exactly one of these two red items. 2 pts

4. A student adds "red shirt" and "red pants" to count the outfits with
less a red article. Correct its result and explain the double counting. 2 pts

Corrected · Common duty · B

Exercise 1 · Discreet evolution

1. [0.5 pt] The relationship is applied successively to rows 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{32}{5} = \frac{34}{5}, \qquad u_2 = \frac{1}{5} \times \frac{34}{5} + \frac{32}{5} = \frac{194}{25}.$$

2. [2 pt] For $n \in \mathbb{N}$,note $P(n)$ property $u_n = 8 - 6\left(\frac{1}{5}\right)^n$.

Initialization. For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. So $P(0)$ is true.

Heredity. Set an integer $n \geq 0$ and suppose $P(n)$ True. By the relationship defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{32}{5} \\ &= \frac{1}{5}\left[8 - 6\left(\frac{1}{5}\right)^n\right] + \frac{32}{5} \\ &= \frac{8}{5} + \frac{32}{5} - 6\left(\frac{1}{5}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. The principle of recurrence gives $u_n = 8 - 6\left(\frac{1}{5}\right)^n$ for every $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{5} < 1$, for everything $n \in \mathbb{N}$ We have $0. < (\frac{1}{5})^n \leq 1$. Multiplication by $-6 < 0$ reverse inequalities:

$$-6 \leq -6(\frac{1}{5})^n < 0.$$

By adding 8 and using the previous formula, you get $2 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6(\frac{1}{5})^{n+1} + 6(\frac{1}{5})^n \\ &= 6(1 - \frac{1}{5})(\frac{1}{5})^n = \frac{24}{5}(\frac{1}{5})^n > 0. \end{aligned}$$

The rest is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 6(\frac{1}{5})^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Calculated and justified limits

1. [1 pt] For $n \geq 1$, we divides by $n^2 \neq 0$:

$$u_n = \frac{4 + \frac{5}{n} - \frac{1}{n^2}}{5 + \frac{2}{n^2}}.$$

Terms $\frac{1}{n}$ and$\frac{1}{n^2}$ The numerator tends to 4 and the denominator to 5 $\neq 0$; by quotient, $u_n \to \frac{4}{5}$.

2. [1 pt] The conjugate is positive. $n \geq 1$, $\sqrt{n^2} = n$, where

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. [1 pt] The denominator tends to 2 ; limit $\frac{5}{2}$.

4. [1 pt] w $_n = n^2(4-\frac{5}{n}) \to +\infty$; the second factor tends to 4 > 0.

5. [1 pt] No: $(2n)-n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

Exercise 3 · One root and two infinites

1. $\sqrt{x^2+4} > |x| \geq x$, therefore $g(x) > 0$.

2. By conjugation $g(x) = 4/(\sqrt{x^2+4}+x)$. The denominator tends to $+\infty$, therefore $g \to 0$. Transformation 1 point, limit 1 point.

3. In $-\infty$, $\sqrt{x^2+4} \to +\infty$ and $-x \to +\infty$, therefore $g \to +\infty$. Writing $\sqrt{x^2} = x$ would be wrong for $x < 0$.

Exercise 4 · Build a decision tree

1. $3 \times 4 \times 2 = 24$.

2. $1 \times 1 \times 2 = 2$.

3. Red shirt only: 1 $\times 3 \times 2 = 6$; red trousers only : 2 $\times 1 \times 2 = 4$. Cases are 10. One point per case.

4. Red shirt: 8 ; red pants: 6. The two outfits with both are counted twice, so $8+6-2 = 12$. Explanation 1 point, result 1 point.

DS3 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{5x+4}{x-1}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 1^- and 1^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim_{x \to +\infty} x^5 e^{-1x}$ justifying. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 + 4x - 5$. Note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (0.5 pt)

3. Demonstrate that the equation $f(x) = 0$ admits a single real solution α. Give its exact value if it is integer; otherwise, frame it between two consecutive integers. (1.5 pt)

4. Study the convexity of f, then determine the co-ordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0, then study the position of the curve in relation to this tangent. (1 pt)

DS3 · Zero 1 · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+3)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Demonstrate that the equation $f(x) = \frac{3}{2}$ admits a single solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Set up a group — 5 points

A group consists of 6 teachers and 10 students, all distinct.
The order of choice does not intervene.

1. How many groups of four people can be trained? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We select a group containing exactly two teachers, then we appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group accompanied of these two functions. (1 pt)
5. Explain why the number of ordered lists of four separate people, divided by 4 !, gives the number of groups obtained in question 1. (1 pt)

Corrected · DS3 · Zero 1

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator cancels only in 1: $D_f = \mathbb{R} \setminus \{1\}$. Like 5 $x + 4 = 5(x-1) + 9$, on obtains $f(x) = 5 + \dfrac{9}{x-1}$.

2. [1 pt] The two limits are worth 5.

3. [1 pt] When $x \to 1^-$, $x - 1 \to 0$ by negative values; as 9 > 0, $\frac{9}{x-1} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 1$ and $y = 5$; curve below for $x < 1$ and above for $x > 1$.

5. [1 pt] The comparative growth of power and the exponential gives directly $x^5 e^{-x} = \frac{x^5}{e^x} \to 0$ when $x \to +\infty$.

Exercise 2 · Variations, convexity and equation

1. [1 pt] f is derivable on $\mathbb{R}$ and $f'(x) = 3x^2 + 4$. Like 3 $x^2 \geq 0$ and $4 > 0$, $f'(x) > 0$ for all x ; f is strictly increasing on $\mathbb{R}$.

2. [0.5 pt] The dominant term is x^3 : limits $-\infty$ and $+\infty$.

3. [1.5 pt] f is a polynomial, so continue on $\mathbb{R}$. Strict growth and limitations $-\infty, +\infty$ allow to apply the corollary of the VIT: there is a single zero α. $f(1) = 0$, so the solution is exactly 1.

4. [1 pt] f $''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, inflection $(0, -5)$.

5. [1 pt] Tangente $y = 4x - 5$; difference x^3, negative to left and positive to right of 0.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 3$; limit 0 per comparative growth.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	3	$\searrow$	0

3. [1 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the subject, $x + 1 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1.5 pt] Continuity, strict decay, $f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly a solution.

5. [1 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 3 - 2x$. A convex function is above each of its tangents, $(x+3)e^{-x} \geq 3 - 2x$ for every $x \geq 0$.

Exercise 4 · Set up a group

1. [1 pt] The order does not distinguish groups: you have to choose a size 4 part out of 16 people, let $\binom{16}{4} = 1820$ groups.

2. [1 pt] Two teachers are selected from among 6 and two students from among 10. a single group: $\binom{6}{2}\binom{10}{2} = 675$.

3. [1 pt] Excluded groups are those trained solely by teachers or only by pupils; These two categories are disjointed. $\binom{16}{4} - \binom{6}{4} - \binom{10}{4} = 1595$.

4. [1 pt] Each group produces 2 $\times$ 2 distributions: 2700.

5. [1 pt] Each group of four individuals corresponds exactly to 4 ! ordered lists. This multiplicity is constant, so division removes multiple counting.

DS3 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{4x+6}{x-2}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)
2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)
3. Determining the limits of f in 2^- and 2^+. (1 pt)
4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)
5. Irrespective of the previous questions, determine $\lim_{x \to +\infty} x^4 e^{-2x}$ justifying. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 + 3x - 7$. Note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)
2. Determine the limits of f in $-\infty$ and in $+\infty$. (0.5 pt)
3. Demonstrate that the equation $f(x) = 0$ admits a single real solution α. Give its exact value if it is integer; otherwise, frame it between two consecutive integers. (1.5 pt)
4. Study the convexity of f, then determine the co-ordinates of the deflection point of $\mathcal{C}_f$. (1 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0, then study the position of the curve in relation to this tangent. (1 pt)

DS3 · Zero 2 · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Demonstrate that the equation $f(x) = 1$ admits a single solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Set up a group — 5 points

A group consists of 9 teachers and 11 students, all separate.
The order of choice does not intervene.

1. How many groups of four people can be trained? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We select a group containing exactly two teachers, then we appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group accompanied of these two functions. (1 pt)
5. Explain why the number of ordered lists of four separate people, divided by 4 !, gives the number of groups obtained in question 1. (1 pt)

Corrected · DS3 · Zero 2

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator cancels only in 2: $D_f = \mathbb{R} \setminus \{2\}$. Like 4 $x + 6 = 4(x-2) + 14$, we get $f(x) = 4 + \dfrac{14}{x-2}$.

2. [1 pt] The two limits are worth 4.

3. [1 pt] When $x \to 2^-$, $x - 2 \to 0$ by negative values; as 14 > 0, $\frac{14}{x-2} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 2$ and $y = 4$; curve below for $x < 2$ and above for $x > 2$.

5. [1 pt] Poisons $t = 2x$. Then $t \to +\infty$ and $x^4 e^{-2x} = \frac{1}{16}\frac{t^4}{e^t}$. Comparative growth gives $\frac{t^4}{e^t} \to 0$; the limit sought therefore is 0.

Exercise 2 · Variations, convexity and equation

1. [1 pt] f is derivable on $\mathbb{R}$ and $f'(x) = 3x^2 + 3$. Like 3 $x^2 \geq 0$ and $3 > 0$, $f'(x) > 0$ for all x ; f is strictly increasing on $\mathbb{R}$.

2. [0.5 pt] The dominant term is x^3 : limits $-\infty$ and $+\infty$.

3. [1.5 pt] f is a polynomial, so continue on $\mathbb{R}$. Strict growth and limitations $-\infty, +\infty$ allow to apply the corollary of the VIT: there is a single zero α. $f(1) = -3 < 0$ and $f(2) = 7 > 0$; therefore $1 < \alpha < 2$.

4. [1 pt] f $''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, inflection $(0, -7)$.

5. [1 pt] Tangente $y = 3x - 7$; difference x^3, negative to left and positive to right of 0.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 2$; limit 0 per comparative growth.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+2)e^{-x} = (-x-1)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	2	$\searrow$	0

3. [1 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-1)e^{-x} = xe^{-x}.$$

On the subject, $x + 0 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1.5 pt] Continuity, strict decay, $f(0) = 2 > 1$ and limit 0 < 1 : exactly a solution.

5. [1 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 2 - x$. A convex function is above each of its tangents, $(x+2)e^{-x} \geq 2 - x$ for every $x \geq 0$.

Exercise 4 · Set up a group

1. [1 pt] The order does not distinguish groups: you have to choose a size 4 part out of 20 people, let $\binom{20}{4} = 4845$ groups.

2. [1 pt] Two teachers from 9 and two students from 11 are selected. Each couple of choices gives a single group: $\binom{9}{2}\binom{11}{2} = 1980$.

3. [1 pt] Excluded groups are those trained solely by teachers or only by pupils; These two categories are disjointed. $\binom{20}{4} - \binom{9}{4} - \binom{11}{4} = 4389$.

4. [1 pt] Each group produces 2 $\times$ 2 distributions: 7920.

5. [1 pt] Each group of four individuals corresponds exactly to 4 ! ordered lists. This multiplicity is constant, so division removes multiple counting.

DS3 · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{3x+8}{x-3}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 3^- and 3^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^3 e^{-3x}$ justifying. (1 pt)

Exercise 2 · Distinguish existence and uniqueness — 4 points

On $\mathbb{R}$, $f(x) = x^3 - 3x$.

1. Draw up the table of changes in f. (1 pt)

2. Show that $f(x) = 1$ has exactly three real solutions, giving three intervals disjointed men who contain them. 2 pts

3. Identify the bending point of the curve by justifying the change in convexity. (1 pt)

DS3 · A · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+5)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. Calculate $f''(x)$, then study the convexity of f. (1 pt)

4. Demonstrate that the equation $f(x) = \frac{5}{2}$ admits a single solution α in $[0;+\infty[$. (1.5 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Two counts of one group — 6 points

A club consists of 6 senior students and 4 first students, all distinct.
three without assigning a function.

1. Count all groups. (1 pt)

2. Count those with exactly two terminals. (1 pt)

3. Count those with at least one student in each level, in two ways. 2 pts

4. Among these mixed groups, how many teams can be formed with a senior manager?
? Explain why to multiply evenly by 3 would be false. 2 pts

Corrected · DS3 · A

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator cancels only in 3 : $D_f = \mathbb{R} \setminus \{3\}$. Like 3 $x + 8 = 3(x-3) + 17$, we get $f(x) = 3 + \dfrac{17}{x-3}$.

2. [1 pt] The two limits are worth 3.

3. [1 pt] When $x \to 3^-$, $x - 3 \to 0$ by negative values; as 17 > 0, $\frac{17}{x-3} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 3$ and $y = 3$; curve below for $x < 3$ and above for $x > 3$.

5. [1 pt] Poisons $t = 3x$. Then $t \to +\infty$ and $x^3 e^{-3x} = \frac{1}{27}\frac{t^3}{e^t}$. Comparative growth gives $\frac{t^3}{e^t} \to 0$; the limit sought therefore is 0.

Exercise 2 · Distinguish existence and uniqueness

1. $f' = 3(x-1)(x+1)$; growth on $]-\infty;-1]$, decline on $[-1;1]$, growth on $[1;+\infty[$. $f(-1) = 2$, $f(1) = -2$.

2. Polynomial continuity and strict monotonicity on each branch. Values: $f(-2) = -2$, $f(-1) = 2$, $f(0) = 0$, $f(1) = -2$, $f(2) = 2$. The VIT provides a single solution in each of the $]-2;-1[$, $]-1;0[$, $]1;2[$. The three branches cover $\mathbb{R}$ and each have only one solution. Existence 1 point, Singleness and completeness 1 point.

3. $f'' = 6x$ change from negative to positive in 0. The curve changes from concave to convex in $(0;0)$.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 5$; limit 0 per comparative growth.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	5	$\searrow$	0

3. [1 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the subject, $x + 3 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1.5 pt] Continuity, strict decay, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly a solution.

5. [1 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function is above each of its tangents, $(x+5)e^{-x} \geq 5 - 4x$ for every $x \geq 0$.

Exercise 4 · Two counts of one group

1. $\binom{10}{3} = 120$.
2. $\binom{6}{2}\binom{4}{1} = 60$.
3. Supplement: 120 $- \binom{6}{3} - \binom{4}{3} = 96$. Disjunct cases: $\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$. One point by method.
4. With two terminals: 60 $\times 2 = 120$; with a terminal: 36 $\times 1 = 36$, total 156. must be terminal: the number of possibilities depends on the composition. Count 1 point, explanation 1 point.

DS3 · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{2x+5}{x-1}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 1^- and 1^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim_{x \to +\infty} x^2 e^{-1x}$ justifying. (1 pt)

Exercise 2 · Read a derivative before calculating — 4 points

We give $F(x) = x + 2e^{-x}$ on $\mathbb{R}$ and $F'(x) = 1 - 2e^{-x}$.

1. Without logarithm, prove that F' absolve in a single real $a \in]0;1[$. We can use $e > 2$. (2 pts)

2. Deduce variations and convexity of F. (1 pt)

3. Can a horizontal line cut this curve into three distinct points? Justify. (1 pt)

DS3 · B · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+4)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Demonstrate that the equation $f(x) = 2$ admits a single solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Selection subject to incompatibility — 6 points

Among 8 separate students, including Lina and Noah, a group of 3 are selected.

1. Count all groups. (1 pt)
2. Count those that contain Lina and Noah. (1 pt)
3. Count those that contain exactly one of the two. 2 pts
4. Only groups containing the two together are prohibited.
   by additional, then separating "none" and "exactly one". 2 pts

Corrected · DS3 · B

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator cancels only in 1: $D_f = \mathbb{R} \setminus \{1\}$. Like 2 $x + 5 = 2(x-1) + 7$, on obtains $f(x) = 2 + \dfrac{7}{x-1}$.

2. [1 pt] The two limits are 2.

3. [1 pt] When $x \to 1^-$, $x - 1 \to 0$ by negative values; as 7 > 0, $\frac{7}{x-1} \to -\infty$, therefore $f(x) \to -\infty$. To the right, the denominator tends to 0 by positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 1$ and $y = 2$; curve below for $x < 1$ and above for $x > 1$.

5. [1 pt] The comparative growth of power and the exponential gives directly $x^2 e^{-x} = \frac{x^2}{e^x} \to 0$ when $x \to +\infty$.

Exercise 2 · Read a derivative before calculating

1. F' is continuous, $(F')' = 2e^{-x} > 0$ therefore strictly increasing. $F'(0) = -1 < 0$ and $F'(1) = 1 - 2/e > 0$. The VIT gives a root in $]0;1[$, the strict growth of $\mathbb{R}$. Assumptions and signs 1 point, existence and uniqueness 1 point.

2. $F' < 0$ before a, $F' > 0$ after: F Decline and then grow. $F'' > 0$ everywhere, so F is convex.

3. No: each strictly monotonous branch meets one horizontal at most once, so at plus two points in total.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 4$; limit 0 per comparative growth.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	4	$\searrow$	0

3. [1 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the subject, $x + 2 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1.5 pt] Continuity, strict decay, $f(0) = 4 > 2$ and limit 0 < 2 : exactly a solution.

5. [1 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function is above each of its tangents, $(x+4)e^{-x} \geq 4 - 3x$ for every $x \geq 0$.

Exercise 4 · Selection subject to incompatibility

1. $\binom{8}{3} = 56$.

2. Choose the third from the other six: 6.

3. Select which of the two is selected, then two of the other six: 2 $\binom{6}{2} = 30$. One point for every choice.

4. Complementary: 56 $- 6 = 50$. Alternative approach: $\binom{6}{3} + 30 = 20 + 30 = 50$. Both cases are One point by method.

White 1 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.25 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (0.75 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 7$. (0.5 pt)

5. Justify that (v_n) converge, then determine its limit. (0.75 pt)

6. Demonstrate that, for all $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. (0.5 pt)

7. Deduct by induction an increase of $7 - v_n$ according to the n. Then determine, by comparison of whole powers, a ranking from which this increase guarantees $7 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function without argument that returns the first integer n such as $7 - v_n \leq 10^{-3}$. Perform the function and compare the rank obtained to the appropriate rank of the previous question. Explain the possible difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0; +\infty[$ by $f(x) = (x + 5)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = \frac{5}{2}$ admits a single solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0; 2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+6\vec{\jmath}+10\vec{k}, \quad \vec{z}=4\vec{\imath}+6\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 6 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 19 times a two-run event called success and failure. The tests are independent and the probability of success is $p=\frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N\geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j\in\{0,\ldots,19\}$, demonstrate the formula of $P(X=j)$ by counting positions possible successes. (0.5 pt)

7. A series of 19 events is accepted if its number of successes is between 1 and 3, bounds Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Corrected · White 1 · Zero 1

Exercise 1 · Recurrence and bounds

1. [0.25 pt] v $_1 = \sqrt{42+0} = \sqrt{42}$ and $v_2 = \sqrt{42+\sqrt{42}}$.

2. [0,75 pt] Note: $P(n)$: « v_n is defined and $0 \leq v_n \leq 7$ ».

   Initialization. $v_0 = 0$ is defined and belongs to $[0,7]$; $P(0)$ is true.

   Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True, then.

   $$0 \leq 42 \leq 42 + v_n \leq 42 + 7 = 49.$$

   The square root is therefore defined and, by growth of this function on $[0,+\infty[$,

   $$0 \leq v_{n+1} = \sqrt{42+v_n} \leq \sqrt{49} = 7.$$

   So $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,7]$.

3. [0,75 pt] For $n \in \mathbb{N}$, posisons $R(n)$: « $v_{n+1} \geq v_n$ ».

   Initialization. $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   Heredity. Fixes $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 42 and then applying growth square root,

   $$v_{n+2} = \sqrt{42+v_{n+1}} \geq \sqrt{42+v_n} = v_{n+1}.$$

   It's $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \geq v_n$ for every $n \geq 0$: (v_n) is increasing.

4. [0.5 pt] Note: $S(n)$: « $v_n < 7$". In row 0, $v_0 = 0 < 7$. Fixes $n \geq 0$ and suppose $S(n)$ True. Since the radicandes are positive, the strict growth of the square root gives

   $$v_{n+1} = \sqrt{42+v_n} < \sqrt{42+7} = \sqrt{49} = 7.$$

   So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n < 7$ for all $n \in \mathbb{N}$.

5. [0,75 pt] The continuation is increasing and increased by 7: the monotonous convergence theorem gives a real limit $\ell \in [0,7]$. The function $x \mapsto \sqrt{42+x}$ is continuous over this interval. limit in the relationship, $\ell = \sqrt{42+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 42 = 0 \quad \Longleftrightarrow \quad (\ell-7)(\ell+6) = 0.$$

   The Root -6 is excluded by $\ell \geq 0$; therefore $\lim v_n = 7$.

6. [0.5 pt] The previous result ensures $7 - v_{n+1} \geq 0$. By multiplying by the conjugate quantity,

   $$\begin{aligned} 7 - v_{n+1} &= \frac{49 - (42+v_n)}{7+\sqrt{42+v_n}} \\ &= \frac{7-v_n}{7+\sqrt{42+v_n}} \leq \frac{7-v_n}{7}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to 7 > 0.

7. [1 pt] Let's show by induction $T(n)$: « $0 \leq 7 - v_n \leq 7^{1-n}$ In row 0, $7 - v_0 = 7 = 7^1$. Fixing $n \geq 0$ and suppose $T(n)$ True. The previous question gives

$$0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7} \leq \frac{7^{1-n}}{7} = 7^{1-(n+1)}.$$

It's $T(n+1)$. By induction, the framework is true at any level. To have $7^{1-n} \leq 10^{-3}$, it is enough of $7^{n-1} \geq 1000$. Gold$^3 = 343 < 1000$ and $7^4 = 2401 \geq 1000$. Rank 5 is therefore the first guaranteed by this increase, without recourse to logarithm.

8. [0.5 pt]

```
From math import sqrt

def first_rang():
    n, v = 0, 0.0
    while 7-v > 1e-3:
        v = sqrt(42+v)
        n += 1
    return n
```

Convergence to 7 ensures a stop. The loop examines rows in the order: it returns 4, the first ranking meeting the criterion for calculated values. Here, the deviations in rows 3 and 4 are valid about 0 ,00265684 and 0,00018978. They are clearly on both sides of 10^{-3}. Guaranteed rank 5 can be larger; it comes from an increase, not from the exact error.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 5$; limit 0 per comparative growth.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	5	↘	0

3. [0.5 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the subject, $x + 3 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1 pt] Continuity, strict decay, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly a solution.

5. [0.5 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function is above each of its tangents, $(x+5)e^{-x} \geq 5 - 4x$ for every $x \geq 0$.

6. [0.5 pt] We know. $f(0) = 5 > \frac{5}{2}$. In addition, $f(2) = 7e^{-2} \simeq 0{,}9473 < \frac{5}{2}$. Zero of $f - \frac{5}{2}$ is therefore in $]0,2[$. A programme is:

```
From math import exp

def framework():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+5)*exp(-m) > 5/2:
            a = m
        Else :
            b = m
    return a, b
```

The decline in the number of f keeps the frame at each step. The result has a length at most 10^{-3} ; its environment has a mistake at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps are enough.

8. [1 pt] For $x \geq 0$, $x + 5 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a limit value achieved.

Exercise 3 · Vector calculation in a base

1. [0.5 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 4 $\vec{u} + 6\vec{v} = 4(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 6\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. [0.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 4, \qquad \mu = 6, \qquad \lambda + \mu = 11.$$

The first two equalitys give $\lambda + \mu = 10$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 4\vec{v} = 6(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 4\vec{\jmath} + 2\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The formula of the midpoint gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 4 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

Here $4 \neq 6$: there is no such λ, therefore A, I, D are not aligned.

8. [1 pt] No: D belongs to the plan but E does not belong to it, based on the demonstrated non-coplanarity. The intersection is exactly D.

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 19 independent identical tests with two exits; $X \sim \mathcal{B}(19, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{19}{2}(\frac{3}{10})^2(\frac{7}{10})^{17} \simeq 0{,}035802$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{19} - 19(\frac{3}{10})(\frac{7}{10})^{18} \simeq 0{,}989578$.

4. [1 pt] E $(X) = \frac{57}{10}$, $V(X) = \frac{399}{100}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1-p)^N$. Probability is therefore 1 $-(\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. Calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 19$. For $j \in \{0, \dots, n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1-p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] Events $X = 1$, $X = 2$, $X = 3$ are incompatible, therefore

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{19}{j} (\frac{3}{10})^j (\frac{7}{10})^{19-j} \simeq 0{,}132031.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 19, 3/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 9. The cumulations in rows 8 and 9 are approximately 0.916085 and 0.967447: this confirms the minimality.

White 1 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.25 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (0.75 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 11$. (0.5 pt)

5. Justify that (v_n) converge, then determine its limit. (0.75 pt)

6. Demonstrate that, for all $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. (0.5 pt)

7. To deduct by induction an increase of $11 - v_n$ according to the n. Then determine, by comparison of entire powers, a ranking from which this increase guarantees $11 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function without argument that returns the first integer n such as $11 - v_n \leq 10^{-3}$. Perform the function and compare the rank obtained to the appropriate rank of the previous question. Explain the possible difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0; +\infty[$ by $f(x) = (x + 4)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = 2$ admits a single solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0; 2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+8\vec{\jmath}+11\vec{k}, \quad \vec{z}=3\vec{\imath}+8\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 8 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 16 times a two-run test called success and failure. The tests are independent and the probability of success is $p=\frac{1}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N\geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j\in\{0,\dots,16\}$, demonstrate the formula of $P(X=j)$ by counting positions possible successes. (0.5 pt)

7. A series of 16 events is accepted if the number of hits is between 1 and 3 points Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Corrected · White 1 · Zero 2

Exercise 1 · Recurrence and bounds

1. [0.25 pt] v $_1 = \sqrt{110+0} = \sqrt{110}$ and $v_2 = \sqrt{110+\sqrt{110}}$.

2. [0,75 pt] Note: $P(n)$: « v_n is defined and $0 \leq v_n \leq 11$ ».

   Initialization. $v_0 = 0$ is defined and belongs to $[0,11]$; $P(0)$ is true.

   Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True, then.

   $$0 \leq 110 \leq 110 + v_n \leq 110 + 11 = 121.$$

   The square root is therefore defined and, by growth of this function on $[0,+\infty[$,

   $$0 \leq v_{n+1} = \sqrt{110+v_n} \leq \sqrt{121} = 11.$$

   So $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,11]$.

3. [0,75 pt] For $n \in \mathbb{N}$, posisons $R(n)$: « $v_{n+1} \geq v_n$ ».

   Initialization. $v_1 = \sqrt{110} > 0 = v_0$; $R(0)$ is true.

   Heredity. Fixes $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 110 and then applying growth square root,

   $$v_{n+2} = \sqrt{110+v_{n+1}} \geq \sqrt{110+v_n} = v_{n+1}.$$

   It's $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \geq v_n$ for every $n \geq 0$: (v_n) is increasing.

4. [0.5 pt] Note: $S(n)$: « $v_n < 11$". In row 0, $v_0 = 0 < 11$. Fixes $n \geq 0$ and suppose $S(n)$ True. Since the radicandes are positive, the strict growth of the square root gives

   $$v_{n+1} = \sqrt{110+v_n} < \sqrt{110+11} = \sqrt{121} = 11.$$

   So $S(n+1)$ is true. Initialization and inheritance establish by induction that $v_n < 11$ for all $n \in \mathbb{N}$.

5. [0,75 pt] The continuation is increasing and increased by 11: the monotonous convergence theorem gives a real limit $\ell \in [0,11]$. The function $x \mapsto \sqrt{110+x}$ is continuous over this interval. the limit in the relationship, $\ell = \sqrt{110+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 110 = 0 \iff (\ell-11)(\ell+10) = 0.$$

   The Root -10 is excluded by $\ell \geq 0$; therefore $\lim v_n = 11$.

6. [0.5 pt] The previous result ensures $11 - v_{n+1} \geq 0$. By multiplying by the conjugate quantity,

   $$\begin{aligned} 11 - v_{n+1} &= \frac{121 - (110+v_n)}{11+\sqrt{110+v_n}} \\ &= \frac{11-v_n}{11+\sqrt{110+v_n}} \leq \frac{11-v_n}{11}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to 11 > 0.

7. [1 pt] Let's show by induction $T(n)$: « $0 \leq 11 - v_n \leq 11^{1-n}$ In row 0, 11 $- v_0 = 11 = 11^1$. Fixes $n \geq 0$ and suppose $T(n)$ True. The previous question gives

$$0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11} \leq \frac{11^{1-n}}{11} = 11^{1-(n+1)}.$$

It's $T(n+1)$. By induction, the framework is true at any level. To have 11 $^{1-n} \leq 10^{-3}$, it just 11 $^{n-1} \geq 1000$. Gold $11^2 = 121 < 1000$ and $11^3 = 1331 \geq 1000$. Rank 4 is therefore the first guaranteed by this increase, without the use of logarithm.

8. [0.5 pt]

```
From math import sqrt

def first_rang():
    n, v = 0, 0.0
    while 11-v > 1e-3:
        v = sqrt(110+v)
        n += 1
    return n
```

The convergence towards 11 ensures the stop. The loop examines the rows in the order: it returns 4, the first ranking meeting the criterion for calculated values. Here, the deviations in rows 3 and 4 are valid about 0 ,00105884 and 0,00004813. They are clearly on both sides of 10 $^{-3}$. Guaranteed rank 4 can be larger; it comes from an increase, not from the exact error.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 4$; limit 0 per comparative growth.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	4	$\searrow$	0

3. [0.5 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the subject, $x + 2 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1 pt] Continuity, strict decay, $f(0) = 4 > 2$ and limit $0 < 2$: exactly a solution.

5. [0.5 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function is above each of its tangents, $(x+4)e^{-x} \geq 4 - 3x$ for every $x \geq 0$.

6. [0.5 pt] We know. $f(0) = 4 > 2$. In addition, $f(2) = 6e^{-2} \simeq 0{,}8120 < 2$. The zero of $f - 2$ is therefore in $]0,2[$. A programme is:

```
From math import exp

def framework():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+4)*exp(-m) > 4/2:
            a = m
        Else :
            b = m
    return a, b
```

The decline in the number of f keeps the frame at each step. The result has a length at most 10^{-3} ; its environment has a mistake at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps are enough.

8. [1 pt] For $x \geq 0$, $x + 4 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a limit value achieved.

Exercise 3 · Vector calculation in a base

1. [0.5 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 3 $\vec{u} + 8\vec{v} = 3(\vec{\imath} + \vec{k}) + 8(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 8\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [0.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 3, \qquad \mu = 8, \qquad \lambda + \mu = 12.$$

The first two equalitys give $\lambda + \mu = 11$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $8\vec{u} - 3\vec{v} = 8(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 8\vec{\imath} - 3\vec{\jmath} + 5\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The formula of the midpoint gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 3 = \frac{\lambda}{2} \text{ and } 8 = \frac{\lambda}{2}.$$

Here $3 \neq 8$: there is no such λ, therefore A, I, D are not aligned.

8. [1 pt] No: D belongs to the plan but E does not belong to it, based on the demonstrated non-coplanarity. The intersection is exactly D.

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 16 independent identical tests with two exits; $X \sim \mathcal{B}(16, \frac{1}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{16}{2}(\frac{1}{10})^2(\frac{9}{10})^{14} \simeq 0{,}274522$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{16} - 16(\frac{1}{10})(\frac{9}{10})^{15} \simeq 0{,}485272$.

4. [1 pt] E $(X) = \frac{8}{5}$, $V(X) = \frac{36}{25}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1-p)^N$. Probability is therefore 1 $-(\frac{9}{10})^N$. It is strictly increasing with N, because $0 < \frac{9}{10} < 1$. Calculations give

$$P_{28} \simeq 0{,}947665 < 0{,}95, \qquad P_{29} \simeq 0{,}952899 \geq 0{,}95.$$

Thus 29 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 16$. For $j \in \{0, \dots, n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1-p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$

7. [0.5 pt] Events $X = 1$, $X = 2$, $X = 3$ are incompatible, therefore

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{16}{j} (\frac{1}{10})^j (\frac{9}{10})^{16-j} \simeq 0{,}746292.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 16, 1/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 4. The cumulations in rows 3 and 4 are about 0.931594 and 0.982996: this confirms the minimality.

White 1 · A

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Two reasonings on a continuation — 4 points

We define $u_0 = 1$, $u_{n+1} = \sqrt{2 + u_n}$.

1. Demonstrate by induction $1 \leq u_n < 2$. (1 pt)

2. Show that $u_{n+1} \geq u_n$, then the sequence converges. (1 pt)

3. Determine its limit by indicating the hypothesis of the transition to the limit. (1 pt)

4. Show $0 \leq 2 - u_{n+1} \leq \frac{1}{3}(2 - u_n)$; · explain the advantage of this information on a simple Conjecture. (1 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0; +\infty[$ by $f(x) = (x + 3)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = \frac{3}{2}$ admits a single solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0; 2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · A · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+5\vec{\jmath}+7\vec{k}, \quad \vec{z}=2\vec{\imath}+5\vec{\jmath}+8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 5 $\vec{u}-2\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Two workshops and one order — 6 points

A component comes from A with probability 0 ,6, if not B. The default rates are respectively 0,02 and 0,05. The components of a control are assumed to be independent and derived from the same production.

1. Build the tree and determine probability p a defect. (1 pt)

2. A component is defective. Calculate the probability that it comes from B. (1 pt)

3. For 20 components, define a binomial variable and calculate the probability of at least one defect. (1 pt)

4. Determine the expected number of defects. Interpret. (1 pt)

5. Write a Python function giving the smallest order size for which probability at least one defect exceeds 0.5, without logarithm. Justify stop. 2 pts

Corrected · White 1 · A

Exercise 1 · Two reasonings on a continuation

1. $u_0 = 1 \in [1;2[$. If $1 \leq u_n < 2$, root growth gives $\sqrt{3} \leq u_{n+1} < 2$, whereinininvariant to Any row.

2. $u_{n+1}^2 - u_n^2 = 2 + u_n - u_n^2 = (2 - u_n)(u_n + 1) > 0$. The two terms are therefore positive $u_{n+1} > u_n$; growth and increase implies convergence.

3. Out of continuity of $x \mapsto \sqrt{2+x}$ on $[1;2]$, $\ell = \sqrt{2+\ell}$. So $(\ell - 2)(\ell + 1) = 0$; $\ell \geq 1$ requires $\ell = 2$.

4. Conjugation: $2 - u_{n+1} = (2 - u_n)/(2 + \sqrt{2+u_n})$. The denominator is at least 3, hence the boundary. It controls error in all ranks, not only on the first calculated.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 3$; limit 0 per comparative growth.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	3	$\searrow$	0

3. [0.5 pt] By deriving the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the subject, $x + 1 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1 pt] Continuity, strict decay, $f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly a solution.

5. [0.5 pt] The tangent in 0 has for equation $y = f(0) + f'(0)x = 3 - 2x$. A convex function is above each of its tangents, $(x+3)e^{-x} \geq 3 - 2x$ for every $x \geq 0$.

6. [0.5 pt] We know. $f(0) = 3 > \frac{3}{2}$. In addition, $f(2) = 5e^{-2} \simeq 0{,}6767 < \frac{3}{2}$. Zero of $f - \frac{3}{2}$ is therefore in $]0,2[$. A programme is:

```
From math import exp

def framework():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+3)*exp(-m) > 3/2:
            a = m
        Else :
            b = m
    return a, b
```

The decline in the number of f keeps the frame at each step. The result has a length at most 10^{-3} ; its environment has a mistake at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps are enough.

8. [1 pt] For $x \geq 0$, $x + 3 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a limit value achieved.

Exercise 3 · Vector calculation in a base

1. [0.5 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 2 $\vec{u} + 5\vec{v} = 2(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 2\vec{\imath} + 5\vec{\jmath} + 7\vec{k} = \vec{w}$.

3. [0.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 2, \qquad \mu = 5, \qquad \lambda + \mu = 8.$$

The first two equalitys give $\lambda + \mu = 7$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $5\vec{u} - 2\vec{v} = 5(\vec{\imath} + \vec{k}) - 2(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 2\vec{\jmath} + 3\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The formula of the midpoint gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 2 = \frac{\lambda}{2} \text{ and } 5 = \frac{\lambda}{2}.$$

Here $2 \neq 5$: there is no such λ, therefore A, I, D are not aligned.

8. [1 pt] No: D belongs to the plan but E does not belong to it, based on the demonstrated non-coplanarity. The intersection is exactly D.

Exercise 4 · Two workshops and one order

1. Tree: A/0 ,6 then default/0 ,02 ; B/0,4 then default/0 ,05, with the supplementarys. $p = 0{,}6(0{,}02) + 0{,}4(0{,}05) = 0{,}032$.

2. $P_D(B) = 0{,}4(0{,}05)/0{,}032 = 0{,}625$.

3. X account for defects: $X \sim \mathcal{B}(20; 0{,}032)$. $P(X \geq 1) = 1 - 0{,}968^{20}$.

4. $E(X) = 20(0{,}032) = 0.64$ default per command on average on many commands; this is not a guaranteed number in each.

5. Programme (1 pt):

```
def Size():
    n, none = 0, 1.0
    while 1 - none <= 0.5:
        n += 1
        none *= 0.968
    return n
```

$0{,}968^n \to 0$, so the threshold is crossed; its decrease guarantees the first ranking (1 pt). Result: 22 ; control rows 21 and 22.

White 1 · B

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A calculation that never achieves its objective — 4 points

For $n \in \mathbb{N}$, we define $v_0 = 0$, $v_{n+1} = (v_n + 3)/2$.

1. Show by induction $v_n = 3 - 3(1/2)^n$. (1 pt)

2. Determine monotonicity and limit. (1 pt)

3. A program repeats the update as long as $v < 3$. Why does he never finish in arithmetic Exactly? (1 pt)

4. Propose a stop condition that guarantees $|v_n - 3| < 10^{-3}$ and determine the first ranking that It's okay. (1 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = 1$ admits a single solution α in $[0;+\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0;2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · B · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=5\vec{\imath}+6\vec{\jmath}+11\vec{k}, \quad \vec{z}=5\vec{\imath}+6\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 6 $\vec{u}-5\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Reconstitute an unknown probability — 6 points

A two-way experiment is repeated independently three times, with the probability of success $p\in[0;1]$ constant. X the number of successes and $q(p)=P(X\geq 2)$.

1. Justify the model and establish $q(p)=3p^2-2p^3$. (1 pt)

2. Study q on $[0;1]$. (1 pt)

3. Demonstrate that $q(p)=0{,}8$ has a single solution and frame between two hundred consecutives. 2 pts

4. If $p=0.8$, calculate $E(X)$ and $P(X=0)$. (1 pt)

5. Another protocol stops at the first success. Does the number of successes still follow? $\mathcal{B}(3;p)$? Justify for 0 $<p<1$. (1 pt)

Corrected · White 1 · B

Exercise 1 · A calculation that never achieves its objective

1. Initialization: 3 $-3=0$. If the formula is in row n fixed, $(v_n+3)/2 = 3-3(1/2)^{n+1}$. By Recurrence is worth every row.

2. $v_{n+1}-v_n=\frac{3}{2}(1/2)^n>0$; $v_n\to 3$ since $(1/2)^n\to 0$.

3. The formula gives $v_n<3$ for the integer: the threshold is never reached. computer can change this behavior, without contradicting the evidence.

4. Continue until 3 $-v\geq 10^{-3}$. The error is $3/2^n$; $2^{11}=2048<3000<4096=2^{12}$, therefore first row 12.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0)=2$; limit 0 per comparative growth.

2. [0.5 pt] By the product rule,

$$f'(x)=e^{-x}-(x+2)e^{-x}=(-x-1)e^{-x}.$$

The two factors have, respectively, a strictly negative and positive sign on the $[0,+\infty[$; therefore $f'<0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	2	$\searrow$	0

3. [0.5 pt] By deriving the previous product,

$$f''(x)=-e^{-x}-(-x-1)e^{-x}=xe^{-x}.$$

On the subject, $x+0\geq 0$ and $e^{-x}>0$: f is convex. There is no change of convexity in inside the domain, so no point of inflection.

4. [1 pt] Continuity, strict decay, $f(0)=2>1$ and limit $0<1$: exactly a solution.

5. [0.5 pt] The tangent in 0 has for equation $y=f(0)+f'(0)x=2-x$. A convex function is above each of its tangents, $(x+2)e^{-x}\geq 2-x$ for every $x\geq 0$.

6. [0.5 pt] We know. $f(0) = 2 > 1$. In addition, $f(2) = 4e^{-2} \simeq 0{,}5413 < 1$. The zero of $f - 1$ is therefore in $]0,2[$. A programme is:

```
From math import exp

def framework():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+2)*exp(-m) > 2/2:
            a = m
        Else :
            b = m
    return a, b
```

The decline in the number of f keeps the frame at each step. The result has a length at most 10^{-3} ; its environment has a mistake at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps are enough.

8. [1 pt] For $x \geq 0$, $x + 2 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a limit value achieved.

Exercise 3 · Vector calculation in a base

1. [0.5 pt] Equal $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, contradiction by the uniqueness of the coordinates in a base.

2. [0.5 pt] By developing, 5 $\vec{u} + 6\vec{v} = 5(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 5\vec{\imath} + 6\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [0.5 pt] Like $\vec{u}, \vec{v}$ are not colinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of decomposition in the base requires

$$\lambda = 5, \qquad \mu = 6, \qquad \lambda + \mu = 12.$$

The first two equalitys give $\lambda + \mu = 11$, in contradiction with the third. vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 5\vec{v} = 6(\vec{\imath} + \vec{k}) - 5(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 5\vec{\jmath} + \vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-colonial and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The formula of the midpoint gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 5 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

Here $5 \neq 6$: there is no such λ, therefore A, I, D are not aligned.

8. [1 pt] No: D belongs to the plan but E does not belong to it, based on the demonstrated non-coplanarity. The intersection is exactly D.

Exercise 4 · Reconstitute an unknown probability

1. $X \sim \mathcal{B}(3; p)$ by independence, same probability and fixed number of three tests. $q = \binom{3}{2}p^2(1 - p) + p^3 = 3p^2 - 2p^3$.

2. $q' = 6p(1-p) > 0$ on $]0;1[$; strict growth from 0 to 1.

3. Polynomial continuity, strict growth and $0 < 0{,}8 < 1$ give existence and uniqueness (1 pt).
$q(0{,}71) = 0{,}796478 < 0{,}8 < q(0{,}72) = 0.808704$, therefore solution in $]0{,}71;0{,}72[$ (1 pt).

4. $E(X) = 2{,}4$; $P(X=0) = 0{,}2^3 = 0{,}008$.

5. No: at most one success is observed with a halt to the first; the binomial distribution would give a probability strictly positive to two and three successes.

DS4 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)
5. Demonstrate that the equation $f(x)=\frac{1}{2}e^2$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 2x^2-6x+2$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)
4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^2$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · Zero 1 · continued

Exercise 3 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0; +\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 13$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 9$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(11; 0; 0)$, $B(0; 11; 0)$ and $C(0; 0; 11)$. We give $\vec{n} = (1; 1; 1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Corrected · DS4 · Zero 1

Exercise 1 · A logarithmic function

1. [1 pt] In 0 $^+$, the numerator tends to $-\infty$ and denominator to 0 $^+$: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3 + \ln x)}{x^2} = \frac{-2 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-2}$, is positive for 0 $< x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is only reached in $s = e^{-2}$ and is

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Like $x > 0$, $f(x) = 0$ equals 3 $+ \ln x = 0$, either $x = e^{-3}$.

4. [1 pt] We're drifting. $(-2 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2 - \ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator cancels in $t = e^{-\frac{3}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Poisons $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

Exercise 2 · Primitives and imposed condition

1. [1 pt] By linearity, antiderivatives are $F(x) = \frac{2}{3}x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 2x^2 - 6x + 2$. The derivative does not depend on C, and all antiderivatives differ from a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{2x}}{} + C$. The condition $F(0) = 6$ requires $\frac{1}{} + C = 6$, where $C = \frac{11}{2}$. The derivative of $\frac{e^{2x}}{2}$ is e^{2x} : the function obtained checks both conditions well.

3. [1 pt] Poisons $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 6) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. [1 pt] A antiderivative is $F(x) = \frac{(x^2+2)^3}{3}$. Indeed, by derivation of a compound,

$$F'(x) = \frac{3}{3}2x(x^2+2)^2 = 2x(x^2+2)^2.$$

5. [1 pt] The derivative of their difference is zero over the interval; the difference is therefore constant. character of interval intervenes in this result.

Exercise 3 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{5}y + \frac{8}{5}$ gives $y = 8$.

2. [1 pt] The constant solution is 8. $z = y - 8$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are: $z(t) = Ce^{-\frac{t}{5}}$, so all the solutions of the initial equation are $y(t) = 8 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 13$ gives $C = 5$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{8}{5}, \qquad y(0) = 13.$$

   The solution of the initial problem is therefore $y(t) = 8 + 5e^{-\frac{t}{5}}$.

3. [1 pt] y $' = -e^{-\frac{t}{5}} < 0$; limit 8.

4. [1 pt] Like $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{5}}$ is negative: y is strictly decreasing.
   Solve

$$\begin{aligned} y(t) \leq 9 &\iff 5e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 5 \\ &\iff t \geq 5\ln 5. \end{aligned}$$

   The last step reverses inequality because we multiply by $-5 < 0$. The first moment is 5 $\ln 5 \simeq$ 8,0472.

5. [1 pt] Y $= 13$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{8}{5})$, $t_j = jh$.

Exercise 4 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-11, 11, 0)$ and $\overrightarrow{AC} = (-11, 0, 11)$ are non-collinear and their scalar products with $\vec{n}$ They suck.

2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1, 1, 1)$. Its point H written so (t, t, t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 11) + t + t = 3t - 11 = 0.$$

   So $t = \frac{11}{3}$ and $H = (\frac{11}{3}, \frac{11}{3}, \frac{11}{3})$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it is Well, the wanted project.

3. [1 pt] $OH = \frac{11}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 11\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(11\sqrt{2})^2 - (\frac{11\sqrt{2}}{2})^2} = \frac{11\sqrt{6}}{2}.$$

   Its area is therefore worth $\frac{1}{2} \times 11\sqrt{2} \times \frac{11\sqrt{6}}{2} = \frac{121}{2}\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(\frac{121}{2}\sqrt{3})(\frac{11}{\sqrt{3}})}{3} = \frac{1331}{6}$. Base OAB : area $\frac{121}{2}$ and height 11, same value.

DS4 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)
5. Demonstrate that the equation $f(x)=\frac{1}{2}e^1$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 5x^2-7x+3$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0)=7$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+7}$. (1 pt)
4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+3)^5$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · Zero 2 · continued

Exercise 3 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0; +\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 16$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 11$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(7;0;0)$, $B(0;7;0)$ and $C(0;0;7)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Corrected · DS4 · Zero 2

Exercise 1 · A logarithmic function

1. [1 pt] In 0 $^+$, the numerator tends to $-\infty$ and denominator to 0 $^+$: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2 + \ln x)}{x^2} = \frac{-1 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-1}$, is positive for 0 $< x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is only reached in $s = e^{-1}$ and is

$$f(s) = \frac{2 - 1}{e^{-1}} = e^1.$$

Like $x > 0$, $f(x) = 0$ equals 2 $+ \ln x = 0$, either $x = e^{-2}$.

4. [1 pt] We're drifting. $(-1 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1 - \ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator cancels in $t = e^{-\frac{1}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Poisons $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

Exercise 2 · Primitives and imposed condition

1. [1 pt] By linearity, antiderivatives are $F(x) = \frac{5}{3}x^3 - \frac{7}{2}x^2 + 3x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 5x^2 - 7x + 3$. The derivative does not depend on C, and all antiderivatives differ from a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{5x}}{} + C$. The condition $F(0) = 7$ imposes $^1 + C = 7$, where $C = \frac{34}{5}$. The derivative of $\frac{e^{5x}}{5}$ is e^{5x} : the function obtained checks both conditions well.

3. [1 pt] Poisons $u(x) = x^2 + 7 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 7) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+7}$.

4. [1 pt] A antiderivative is $F(x) = \frac{(x^2 + 3)^6}{6}$. Indeed, by derivation of a compound,

$$F'(x) = \frac{6}{6}2x(x^2 + 3)^5 = 2x(x^2 + 3)^5.$$

5. [1 pt] The derivative of their difference is zero over the interval; the difference is therefore constant. character of interval intervenes in this result.

Exercise 3 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{4}y + \frac{5}{2}$ gives $y = 10$.

2. [1 pt] The constant solution is 10. $z = y - 10$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are $z(t) = Ce^{-\frac{t}{4}}$, so all the solutions of the initial equation are $y(t) = 10 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 16$ given $C = 6$. For the function obtained,

$$y'(t) = -\frac{3}{2}e^{-\frac{t}{4}} = -\frac{y(t)}{4} + \frac{5}{2}, \qquad y(0) = 16.$$

The solution of the initial problem is therefore $y(t) = 10 + 6e^{-\frac{t}{4}}$.

3. [1 pt] y $' = -\frac{3}{2}e^{-\frac{t}{4}} < 0$; limit 10.

4. [1 pt] Like $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -\frac{3}{2}e^{-\frac{t}{4}}$ is negative: y is strictly decreasing. Solve

$$\begin{aligned} y(t) \leq 11 &\iff 6e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 6 \\ &\iff t \geq 4\ln 6. \end{aligned}$$

The last step reverses inequality because we multiply by $-4 < 0$. The first moment is 4 $\ln 6 \simeq$ 7,1670.

5. [1 pt] Y $= 16$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + \frac{5}{2})$, $t_j = jh$.

Exercise 4 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-7,7,0)$ and $\overrightarrow{AC} = (-7,0,7)$ are non-collinear and their scalar products with $\vec{n}$ They suck.

2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1,1,1)$. Its point H written so (t,t,t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-7) + t + t = 3t - 7 = 0.$$

So $t = \frac{7}{3}$ and $H = (\frac{7}{3}, \frac{7}{3}, \frac{7}{3})$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it's good the project sought.

3. [1 pt] $OH = \frac{7}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 7\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(7\sqrt{2})^2 - (\frac{7\sqrt{2}}{2})^2} = \frac{7\sqrt{6}}{2}.$$

Its area is therefore worth $\frac{1}{2} \times 7\sqrt{2} \times \frac{7\sqrt{6}}{2} = \frac{49}{2}\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(\frac{49}{2}\sqrt{3})(\frac{7}{\sqrt{3}})}{3} = \frac{343}{6}$. Base OAB : area $\frac{49}{2}$ and height 7, same value.

DS4 · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)
5. Demonstrate that the equation $f(x)=\frac{1}{2}e^4$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 4x^2-4x+1$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{4x}$ on $\mathbb{R}$ such that $F(0)=4$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+4}$. (1 pt)
4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+1)^4$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · A · continued

Exercise 3 · Identify and resolve cooling — 4 points

Temperature $T(t)$ in degrees Celsius, t in minutes, check $T' = -k(T - 20)$ where: $k > 0$. Measurement $T(0) = 80$ and $T(10) = 50$.

1. Express T according to the k. (1 pt)

2. Determine k and specify its unit. (1 pt)

3. Determine the first time that $T(t) \leq 35$ and interpret. 2 pts

Exercise 4 · Nearest straight point — 6 points

In an orthonormal coordinate system, $A(1;0;0)$, $M(0;2;3)$ and d is the right passing through A of Director $u = (1;1;0)$. One point P of d checks $\overrightarrow{AP} = tu$.

1. Express the coordinates of P. (1 pt)

2. Determine the point H of d such that $\overrightarrow{MH} \cdot u = 0$. 2 pts

3. Developing MP^2 and put it in form 2 $(t - \frac{1}{2})^2 + c$. 2 pts

4. Find the minimum distance and compare the two methods. (1 pt)

Corrected · DS4 · A

Exercise 1 · A logarithmic function

1. [1 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5 + \ln x)}{x^2} = \frac{-4 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is only reached in $s = e^{-4}$ and is

$$f(s) = \frac{5 - 4}{e^{-4}} = e^4.$$

Like $x > 0$, $f(x) = 0$ equals $5 + \ln x = 0$, either $x = e^{-5}$.

4. [1 pt] We're drifting. $(-4 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4 - \ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator cancels in $t = e^{-\frac{7}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Poisons $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

Exercise 2 · Primitives and imposed condition

1. [1 pt] By linearity, antiderivatives are $F(x) = \frac{4}{3}x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 4x^2 - 4x + 1$. The derivative does not depend on C, and all antiderivatives differ from a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{4x}}{} + C$. The condition $F(0) = 4$ imposes $^1 + C = 4$, where $C = \frac{15}{4}$. The derivative of $\frac{e^{4x}}{4}$ is e^{4x} : the function obtained checks both conditions well.

3. [1 pt] Poisons $u(x) = x^2 + 4 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 4) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+4}$.

4. [1 pt] A antiderivative is $F(x) = \frac{(x^2+1)^5}{5}$. Indeed, by derivation of a compound,

$$F'(x) = \frac{5}{5}2x(x^2+1)^4 = 2x(x^2+1)^4.$$

5. [1 pt] The derivative of their difference is zero over the interval; the difference is therefore constant. character of interval intervenes in this result.

Exercise 3 · Identify and resolve cooling

1. Lay down $z = T - 20$, then $z' = -kz$ and $z(0) = 60$. Therefore, $T(t) = 20 + 60e^{-kt}$.

2. $50 = 20 + 60e^{-10k}$, therefore $e^{-10k} = 1/2$ and $k = 2/10$ in min $^{-1}$.

3. $T \leq 35 \iff e^{-kt} \leq 1/4 \iff t \geq \ln 4/k = 20$. The temperature is strictly decreasing, since $T' = -60ke^{-kt} < 0$; the first moment is 20 min. Resolution 1 point, first moment and interpretation 1 point.

Exercise 4 · Nearest straight point

1. $P(1+t;t;0)$.

2. $MH = (1+t;t-2;-3)$; $MH \cdot u = 2t - 1$. Therefore $t = 1/2$ and $H(3/2;1/2;0)$. Equation 1 point, resolution 1 item.

3. $MP^2 = (1+t)^2 + (t-2)^2 + 9 = 2t^2 - 2t + 14 = 2(t-1/2)^2 + 27/2$. Development 1 point, square 1 point.

4. The minimum is $\sqrt{27/2} = 3\sqrt{6}/2$, achieved only in $t = 1/2$. Algebraic minimization find the orthogonal project.

DS4 · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{4+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)
5. Demonstrate that the equation $f(x)=\frac{1}{2}e^3$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 3x^2-6x+2$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{3x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)
4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^3$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · B · continued

Exercise 3 · Variable contribution — 4 points

We're looking for a detritable function. y on $[0;+\infty[$ such that $y' = -y + 2t + 1$ and $y(0) = 3$.

1. Determine a particular affine solution $y_p(t) = at + b$. (1 pt)

2. Show that $z = y - y_p$ checks $z' = -z$, then determine y. 2 pts

3. Check the initial condition and explain why y does not tend to a steady equilibrium. (1 pt)

Exercise 4 · A height calculated in two ways — 6 points

In an orthonormal coordinate system, $O(0;0;0)$, $A(6;0;0)$, $B(0;4;0)$, $C(0;0;3)$. The coordinates are in meters.

1. Calculate the volume of $OABC$ with the base OAB. (1 pt)

2. Verify that $n = (2;3;4)$ is normal on the plane (ABC). 2 pts

3. The point H defined by $\overrightarrow{OH} = tn$ belongs to the plan if $\overrightarrow{AH}$ is orthogonal to n. Determine t then OH. 2 pts

4. Deduct from ABC. (1 pt)

Fixed · DS4 · B

Exercise 1 · A logarithmic function

1. [1 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is only reached in $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Like $x > 0$, $f(x) = 0$ equals $4 + \ln x = 0$, either $x = e^{-4}$.

4. [1 pt] We're drifting. $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator cancels in $t = e^{-\frac{5}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Poisons $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

Exercise 2 · Primitives and imposed condition

1. [1 pt] By linearity, antiderivatives are $F(x) = x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 3x^2 - 6x + 2$. The derivative does not depend on C, and all antiderivatives differ from a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{3x}}{} + C$. The condition $F(0) = 6$ requires $\frac{1}{} + C = 6$, where $C = \frac{17}{3}$. The derivative of $\frac{e^{3x}}{3}$ is e^{3x} : the function obtained checks both conditions well.

3. [1 pt] Poisons $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 6) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. [1 pt] A antiderivative is $F(x) = \frac{(x^2 + 2)^4}{4}$. Indeed, by derivation of a compound,

$$F'(x) = \frac{4}{4} 2x(x^2 + 2)^3 = 2x(x^2 + 2)^3.$$

5. [1 pt] The derivative of their difference is zero over the interval; the difference is therefore constant. character of interval intervenes in this result.

Exercise 3 · Variable contribution

1. $a = -at - b + 2t + 1$ at any time requires $a = 2$, $b = -1$; $y_p = 2t - 1$.

2. By subtraction, $z' = -z$ so $z = Ce^{-t}$. $y(0) = 3$ gives $-1 + C = 3$, either $C = 4$. Thus $y = 2t - 1 + 4e^{-t}$. Transformation 1 point, constant 1 point.

3. $y(0) = -1 + 4 = 3$; $y(t) \to +\infty$ Since the term affine diverges and exponential tends to 0.
The second Member State has a variable contribution in terms of t.

Exercise 4 · A height calculated in two ways

1. $\mathcal{A}(OAB) = 6 \times 4/2 = 12$ m^2 ; height $OC = 3$ m, so volume 12 m ^{3}.

2. $AB = (-6;4;0)$, $AC = (-6;0;3)$ are non-collinear. $n \cdot AB = -12 + 12 = 0$, $n \cdot AC = -12 + 12 =$ Test 1 point, non-collinearity and conclusion 1 point.

3. $AH = (2t - 6; 3t; 4t)$, therefore $AH \cdot n = 29t - 12 = 0$. Thus $t = 12/29$ and $OH = (12/29)\sqrt{29} = 12/\sqrt{29}$ m. Condition 1 point, distance 1 point.

4. $12 = \mathcal{A}(ABC)\, OH/3$, so $\mathcal{A}(ABC) = 3\sqrt{29}$ m^2.

DS5 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (1 pt)

2. Using a party integration, calculate I_1. (1 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 9$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · Zero 1 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(9;0;0)$, $B(0;9;0)$ and $C(0;0;9)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Amount of random variables — 5 points

Set $n = 17$. Considered n independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli's law of parameter $p = \frac{1}{5}$. We're laying $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the law followed by S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Installation $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 17 variables are replaced by 17 variables all equal to X_1. Calculate the variance of their average and compare it with that of question 4. why the results differ. (1 pt)

Corrected · DS5 · Zero 1

Exercise 1 · Comprehensive and follow-up

1. [1 pt] The function $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of e^{4x}. Thus $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. [1 pt] Choices $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two continuously derivative derivative functions on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Part-by-part integration gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}. \end{aligned}$$

3. [1 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Like $e^{4x} > 0$, integration on increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The follow-up is therefore positive and decreasing in the broad sense. the same strict: the integrands of I_n and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{4x} \leq e^4$; therefore $0 \leq I_n \leq e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. [1 pt] Apply PPI to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{4x}}{4}$. Like $n+1 \geq 1$, the term in $x = 0$ sucks. Where

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{n+1}{4}I_n. \end{aligned}$$

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{3}y + \frac{8}{3}$ gives $y = 8$.

2. [1 pt] The constant solution is 8. $z = y - 8$; the equation becomes $z' = -\frac{z}{3}$. Its solutions are: $z(t) = Ce^{-\frac{t}{3}}$, so all the solutions of the initial equation are $y(t) = 8 + Ce^{-\frac{t}{3}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -\frac{4}{3}e^{-\frac{t}{3}} = -\frac{y(t)}{3} + \frac{8}{3}, \qquad y(0) = 12.$$

The solution of the initial problem is therefore $y(t) = 8 + 4e^{-\frac{t}{3}}$.

3. [1 pt] y $' = -\frac{4}{3}e^{-\frac{t}{3}} < 0$; limit 8.

4. [1 pt] Like $e^{-\frac{t}{3}} > 0$, the derivative $y'(t) = -\frac{4}{3}e^{-\frac{t}{3}}$ is negative: y is strictly decreasing. Solve

$$\begin{aligned} y(t) \leq 9 &\iff 4e^{-\frac{t}{3}} \leq 1 \\ &\iff -\frac{t}{3} \leq -\ln 4 \\ &\iff t \geq 3\ln 4. \end{aligned}$$

The last step reverses inequality because we multiply by $-3 < 0$. The first moment is 3 $\ln 4 \simeq$ 4,1589.

5. [1 pt] Y $= 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{3} + \frac{8}{3})$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-9,9,0)$ and $\overrightarrow{AC} = (-9,0,9)$ are non-collinear and their scalar products with $\vec{n}$ They suck.

2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1,1,1)$. Its point H written so (t,t,t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-9) + t + t = 3t - 9 = 0.$$

So $t = 3$ and $H = (3,3,3)$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it is well the Projected.

3. [1 pt] $OH = \frac{9}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 9\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(9\sqrt{2})^2 - (\frac{9\sqrt{2}}{2})^2} = \frac{9\sqrt{6}}{2}.$$

Its area is therefore worth $\frac{1}{2} \times 9\sqrt{2} \times \frac{9\sqrt{6}}{2} = \frac{81}{2}\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(\frac{81}{2}\sqrt{3})(\frac{9}{\sqrt{3}})}{3} = \frac{243}{2}$. Base OAB : area $\frac{81}{2}$ and height 9, same value.

Exercise 4 · Amount of random variables

1. [1 pt] S $\sim \mathcal{B}(17, \frac{1}{5})$; $E(S) = \frac{17}{5}$.

2. [1 pt] Each X_i is 0 or 1, so $X_i^2 = X_i$. Thus $E(X_i) = p = \frac{1}{5}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1-p) = \frac{4}{25}.$$

The variables are independent; the additivity of the variance gives

$$V(S) = \sum_{i=1}^{17} V(X_i) = 17p(1-p) = \frac{68}{25}.$$

3. [1 pt] By linearity of expectation and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 9E(S) - 17 = \frac{68}{5}, \qquad V(G) = 81V(S) = \frac{5508}{25}.$$

The constant -17 does not change the variance.

4. [1 pt] E $(M) = \frac{1}{5}$ and $V(M) = \frac{4}{425}$; the variance factor is $\frac{1}{n^2}$.

5. [1 pt] No. The new average is X_1, its variance $\frac{4}{25}$. Variance additivity used previously requires independence.

DS5 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{3x}$.

1. Calculate I_0. (1 pt)

2. Using a party integration, calculate I_1. (1 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^3}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{2}y + 5$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 15$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 11$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · Zero 2 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(13;0;0)$, $B(0;13;0)$ and $C(0;0;13)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Amount of random variables — 5 points

Set $n = 14$. Considered n independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli's law of parameter $p = \frac{1}{4}$. We're laying $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the law followed by S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Installation $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 14 variables are replaced by 14 variables all equal to X_1. Calculate the variance of their average and compare it with that of question 4. why the results differ. (1 pt)

Corrected · DS5 · Zero 2

Exercise 1 · Comprehensive and follow-up

1. [1 pt] The function $x \mapsto \frac{e^{3x}}{3}$ is an antiderivative of e^{3x}. Thus $I_0 = [\frac{e^{3x}}{3}]_0^1 = \frac{e^3-1}{3}$.

2. [1 pt] Choices $u(x) = x$ and $v(x) = \frac{e^{3x}}{3}$, two continuously derivative derivative functions on $[0,1]$. Then $u' = 1$ and $v' = e^{3x}$. Part-by-part integration gives

$$I_1 = \left[\frac{xe^{3x}}{3}\right]_0^1 - \frac{1}{3}\int_0^1 e^{3x}\,\mathrm{d}x$$
$$= \frac{e^3}{3} - \frac{e^3-1}{9} = \frac{2e^3+1}{9}.$$

3. [1 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Like $e^{3x} > 0$, integration on increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The follow-up is therefore positive and decreasing in the broad sense. the same strict: the integrands of I_n and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{3x} \leq e^3$; therefore $0 \leq I_n \leq e^3 \int_0^1 x^n\,\mathrm{d}x = \frac{e^3}{n+1} \to 0$.

5. [1 pt] Apply PPI to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{3x}}{3}$. Like $n+1 \geq 1$, the term in $x = 0$ sucks. Where

$$I_{n+1} = \left[\frac{x^{n+1}e^{3x}}{3}\right]_0^1 - \frac{n+1}{3}\int_0^1 x^n e^{3x}\,\mathrm{d}x$$
$$= \frac{e^3}{3} - \frac{n+1}{3}I_n.$$

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{2}y + 5$ gives $y = 10$.

2. [1 pt] The constant solution is 10. $z = y - 10$; the equation becomes $z' = -\frac{z}{2}$. Its solutions are $z(t) = Ce^{-\frac{t}{2}}$, so all the solutions of the initial equation are $y(t) = 10 + Ce^{-\frac{t}{2}}$. The condition $y(0) = 15$ gives $C = 5$. For the function obtained,

$$y'(t) = -\frac{5}{2}e^{-\frac{t}{2}} = -\frac{y(t)}{2} + 5, \qquad y(0) = 15.$$

The solution of the initial problem is therefore $y(t) = 10 + 5e^{-\frac{t}{2}}$.

3. [1 pt] y $' = -\frac{5}{2}e^{-\frac{t}{2}} < 0$; limit 10.

4. [1 pt] Like $e^{-\frac{t}{2}} > 0$, the derivative $y'(t) = -\frac{5}{2}e^{-\frac{t}{2}}$ is negative: y is strictly decreasing. Solve

$$y(t) \leq 11 \iff 5e^{-\frac{t}{2}} \leq 1$$
$$\iff -\frac{t}{2} \leq -\ln 5$$
$$\iff t \geq 2\ln 5.$$

The last step reverses inequality because we multiply by $-2 < 0$. The first moment is 2 $\ln 5 \simeq$ 3,2189.

5. [1 pt] Y $= 15$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{2} + 5)$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-13, 13, 0)$ and $\overrightarrow{AC} = (-13, 0, 13)$ are non-collinear and their scalar products with $\vec{n}$ They suck.

2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1, 1, 1)$. Its point H written so (t, t, t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 13) + t + t = 3t - 13 = 0.$$

So $t = \frac{13}{3}$ and $H = (\frac{13}{3}, \frac{13}{3}, \frac{13}{3})$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it is Well, the wanted project.

3. [1 pt] $OH = \frac{13}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 13\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(13\sqrt{2})^2 - (\frac{13\sqrt{2}}{2})^2} = \frac{13\sqrt{6}}{2}.$$

Its area is therefore worth $\frac{1}{2} \times 13\sqrt{2} \times \frac{13\sqrt{6}}{2} = \frac{169}{2}\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(\frac{169}{2}\sqrt{3})(\frac{13}{\sqrt{3}})}{3} = \frac{2197}{6}$. Base OAB : area $\frac{169}{2}$ and height 13, same value.

Exercise 4 · Amount of random variables

1. [1 pt] S $\sim \mathcal{B}(14, \frac{1}{4})$; $E(S) = \frac{7}{2}$.

2. [1 pt] Each X_i is 0 or 1, so $X_i^2 = X_i$. Thus $E(X_i) = p = \frac{1}{4}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1 - p) = \frac{3}{16}.$$

The variables are independent; the additivity of the variance gives

$$V(S) = \sum_{i=1}^{14} V(X_i) = 14p(1 - p) = \frac{21}{8}.$$

3. [1 pt] By linearity of expectation and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 6E(S) - 14 = 7, \qquad V(G) = 36V(S) = \frac{189}{2}.$$

The constant -14 does not change the variance.

4. [1 pt] E $(M) = \frac{1}{4}$ and $V(M) = \frac{3}{224}$; the variance factor is $\frac{1}{n^2}$.

5. [1 pt] No. The new average is X_1, its variance $\frac{3}{16}$. Variance additivity used previously requires independence.

DS5 · A

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Choose two integrations per part — 4 points

Set $I = \int_0^1 x^2 e^x \,\mathrm{d}x$.

1. Justify the choice of functions in a first PPI and bring back I to $J = \int_0^1 xe^x \,\mathrm{d}x$. (1 pt)

2. Calculate J then I. 2 pts

3. Check the framework $\frac{1}{3} \leq I \leq \frac{e}{3}$ without calculating antiderivatives. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{5}y + \frac{6}{5}$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 7$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · A · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(10;0;0)$, $B(0;10;0)$ and $C(0;0;10)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · A cost that is not binomial — 6 points

A random cost X is 0, 2 or 5 euros with respective probabilities $0{,}5$, $0{,}3$, $0{,}2$. Invoice $Y = 0{,}8X + 1$. For n independent operations of the same law, we note S_n the sum of invoicing and $M_n = S_n / n$.

1. Calculate $E(X)$ and $E(X^2)$. (1 pt)

2. Deduct from it $V(X)$, $E(Y)$ and $V(Y)$. 2 pts

3. Calculate $E(M_n)$ and $V(M_n)$ indicating where independence takes place. 2 pts

4. What becomes of the standard deviation of M_n if multiply n By four? (1 pt)

Corrected · DS5 · A

Exercise 1 · Choose two integrations per part

1. Take $u = x^2$, $v = e^x$Class $\mathcal{C}^1$ on $[0;1]$. Then $I = [x^2e^x]_0^1 - 2\int_0^1 xe^x\,\mathrm{d}x = e - 2J$.

2. Second PPI with $u = x$, $v = e^x$: $J = [xe^x]_0^1 - \int_0^1 e^x\,\mathrm{d}x = e - (e-1) = 1$. Therefore $I = e - 2$. One point for each justified result.

3. $1 \leq e^x \leq e$ and $x^2 \geq 0$, therefore $x^2 \leq x^2e^x \leq ex^2$. Integrate on $[0;1]$ Gives the bounds.

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{5}y + \frac{6}{5}$ gives $y = 6$.

2. [1 pt] The constant solution is 6. $z = y - 6$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are: $z(t) = Ce^{-\frac{t}{5}}$, so all the solutions of the initial equation are $y(t) = 6 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 12$ gives $C = 6$. For the function obtained,

$$y'(t) = -\frac{6}{5}e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{6}{5}, \qquad y(0) = 12.$$

The solution of the initial problem is therefore $y(t) = 6 + 6e^{-\frac{t}{5}}$.

3. [1 pt] y $' = -\frac{6}{5}e^{-\frac{t}{5}} < 0$; limit 6.

4. [1 pt] Like $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -\frac{6}{5}e^{-\frac{t}{5}}$ is negative: y is strictly decreasing. Solve

$$\begin{aligned} y(t) \leq 7 &\iff 6e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 6 \\ &\iff t \geq 5\ln 6. \end{aligned}$$

The last step reverses inequality because we multiply by $-5 < 0$. The first moment is 5 $\ln 6 \simeq$ 8,9588.

5. [1 pt] Y $= 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{6}{5})$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-10, 10, 0)$ and $\overrightarrow{AC} = (-10, 0, 10)$ are non-collinear and their scalar products with $\vec{n}$ They suck.

2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1,1,1)$. Its point H written so (t,t,t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-10) + t + t = 3t - 10 = 0.$$

So $t = \frac{10}{3}$ and $H = (\frac{10}{3}, \frac{10}{3}, \frac{10}{3})$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it is Well, the wanted project.

3. [1 pt] $OH = \frac{10}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 10\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(10\sqrt{2})^2 - (\frac{10\sqrt{2}}{2})^2} = \frac{10\sqrt{6}}{2}.$$

Its area is therefore worth $\frac{1}{2} \times 10\sqrt{2} \times \frac{10\sqrt{6}}{2} = 50\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(50\sqrt{3})(\frac{10}{\sqrt{3}})}{3} = \frac{500}{3}$. Base OAB : area 50 and height 10, same value.

Exercise 4 · A cost that is not binomial

1. $E(X) = 0{,}6 + 1 = 1{,}6$; $E(X^2) = 1{,}2 + 5 = 6.2$ (0.5 each).

2. $V(X) = 6{,}2 - 1{,}6^2 = 3{,}64$; $E(Y) = 0{,}8(1{,}6) + 1 = 2{,}28$; $V(Y) = 0{,}8^2(3{,}64) = 2{,}3296$. Variance X 1 point, transformations 1 point.

3. Linearity: $E(S_n) = 2{,}28n$, therefore $E(M_n) = 2{,}28$ (1 pt). Independence: $V(S_n) = 2{,}3296n$, therefore $V(M_n) = 2{,}3296/n$ (1 pt).

4. It is divided by 2 because $\sigma(M_n) = \sqrt{2{,}3296/n}$.

DS5 · B

RESERVED FOR ASSESSMENT

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · An integral family without antiderivatives requested — 4 points

For $n \in \mathbb{N}$, $I_n = \int_0^1 \frac{x^n}{1+x}\,\mathrm{d}x$.

1. Calculate I_0 and establish $I_n + I_{n+1} = \frac{1}{n+1}$. (1 pt)

2. Show that (I_n) is decreasing and positive. (1 pt)

3. Provide guidance on I_n which makes it possible to determine its limit. 2 pts

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{4}y + 2$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 9$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · B · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(6;0;0)$, $B(0;6;0)$ and $C(0;0;6)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Detect a hidden addiction — 6 points

X and Y are two Bernoulli variables independent of parameter 1 /2. Installation $S = X + Y$, $Z = 2X$, $U = S/2$ and $W = Z/2$.

1. Give the law of S. (1 pt)

2. Give the law of Z and calculate $E(S), E(Z)$. (1 pt)

3. Calculate $V(S)$ and $V(Z)$. Explain why the two results differ. 2 pts

4. Compare variances of U and W. Which corresponds to an average of two observations independent? 2 pts

Corrected · DS5 · B

Exercise 1 · An integral family without antiderivatives requested

1. $I_0 = [\ln(1+x)]_0^1 = \ln 2$. The sum of the integrands is x^n, therefore $I_n + I_{n+1} = \int_0^1 x^n \, dx = 1/(n+1)$.
2. For $x \in [0;1]$, $0 \leq x^{n+1} \leq x^n$; divide by 1 $+ x > 0$ then integrate gives 0 $\leq I_{n+1} \leq I_n$.
3. $1/2 \leq 1/(1+x) \leq 1$, that's $1\ /[2(n+1)] \leq I_n \leq 1/(n+1)$ (1 pt). The bounds tend towards 0; the gendarmes give $I_n \to 0$ (1 pt).

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{4}y + 2$ given $y = 8$.
2. [1 pt] The constant solution is 8. $z = y - 8$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are: $z(t) = Ce^{-\frac{t}{4}}$, so all the solutions of the initial equation are $y(t) = 8 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{4}} = -\frac{y(t)}{4} + 2, \qquad y(0) = 12.$$

   The solution of the initial problem is therefore $y(t) = 8 + 4e^{-\frac{t}{4}}$.
3. [1 pt] y $' = -e^{-\frac{t}{4}} < 0$; limit 8.
4. [1 pt] Like $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{4}}$ is negative: y is strictly decreasing. Solve

$$\begin{aligned} y(t) \leq 9 &\iff 4e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 4 \\ &\iff t \geq 4\ln 4. \end{aligned}$$

   The last step reverses inequality because we multiply by $-4 < 0$. The first moment is 4 $\ln 4 \simeq$ 5,5452.
5. [1 pt] Y $= 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + 2)$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-6,6,0)$ and $\overrightarrow{AC} = (-6,0,6)$ are non-collinear and their scalar products with $\vec{n}$ They suck.
2. [1 pt] The normal right to the plane passing through O a for Director $\vec{n} = (1,1,1)$. Its point H written so (t,t,t). For H or in (ABC), it is necessary and it is sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-6) + t + t = 3t - 6 = 0.$$

   So $t = 2$ and $H = (2,2,2)$. This point is in the plan and $\overrightarrow{OH}$ is colinear to normal; it is well the Projected.
3. [1 pt] $OH = \frac{6}{\sqrt{3}}$; for everything P of the plan, $OP^2 = OH^2 + HP^2 \geq OH^2$, equality only for $P = H$.
4. [1 pt] Differences in coordinates give $AB = AC = BC = 6\sqrt{2}$. The triangle is equilateral. Its height divides one side in two equal parts and is

$$\sqrt{(6\sqrt{2})^2 - (\frac{6\sqrt{2}}{2})^2} = \frac{6\sqrt{6}}{2}.$$

Its area is therefore worth $\frac{1}{2} \times 6\sqrt{2} \times \frac{6\sqrt{6}}{2} = 18\sqrt{}$Three district units.

5. [1 pt] Base ABC : volume $\frac{(18\sqrt{3})(\frac{6}{\sqrt{3}})}{3} = 36$. Base OAB : area 18 and height 6, same value.

Exercise 4 · Detect a hidden addiction

1. S is 0, 1, 2 with probabilities 1/4, 1/2, 1/4: law $\mathcal{B}(2;1/2)$.
2. Z is 0 or 2 with probabilities 1/2 each. $E(S) = E(Z) = 1$.
3. $V(S) = V(X) + V(Y) = 1/2$ by independence. $V(Z) = 4V(X) = 1$. Add two copies of the same variable does not add up independent variables. Calculations 1 point, explanation 1 point.
4. $V(U) = V(S)/4 = 1/8$ and $V(W) = V(Z)/4 = 1/4$. $U = (X+Y)/2$ is the independent average; $W = X$ 1 point calculations, 1 point interpretation.

White 2 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{5x}$.

1. Calculate I_0. (0.5 pt)

2. Using a party integration, calculate I_1. (0.5 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^5}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for every $n \in \mathbb{N}$. (0.5 pt)

7. We share $[0;1]$ in N intervals of the same length, $N \geq 1$. Write the amounts L_N and U_N of areas of rectangles constructed with values of e^{5x} at left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N ensuring that this framework has a width of not more than 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{4+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^3$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-3) + \ln(x+3) = \ln(27)$ after having specified the set on which this equation is defined. (1 pt)

White 2 · Zero 1 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 10$ and point $M(3;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 20 times a two-run event, called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 20 events is accepted if its number of successes is between 1 and 3, bounds Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Corrected · White 2 · Zero 1

Exercise 1 · Comprehensive and follow-up

1. [0.5 pt] The function $x \mapsto \frac{e^{5x}}{5}$ is an antiderivative of e^{5x}. Thus $I_0 = [\frac{e^{5x}}{5}]_0^1 = \frac{e^5-1}{5}$.

2. [0.5 pt] Choices $u(x) = x$ and $v(x) = \frac{e^{5x}}{5}$, two continuously derivative derivative functions on $[0,1]$. Then $u' = 1$ and $v' = e^{5x}$. Part-by-part integration gives

$$I_1 = \left[\frac{xe^{5x}}{5}\right]_0^1 - \frac{1}{5}\int_0^1 e^{5x}\,\mathrm{d}x$$
$$= \frac{e^5}{5} - \frac{e^5-1}{25} = \frac{4e^5+1}{25}.$$

3. [0.5 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Like $e^{5x} > 0$, integration on increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The follow-up is therefore positive and decreasing in the broad sense. the same strict: the integrands of I_n and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{5x} \leq e^5$; therefore $0 \leq I_n \leq e^5 \int_0^1 x^n\,\mathrm{d}x = \frac{e^5}{n+1} \to 0$.

5. [0.5 pt] Apply PPI to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{5x}}{5}$. Like $n+1 \geq 1$, the term in $x = 0$ sucks. Where

$$I_{n+1} = \left[\frac{x^{n+1}e^{5x}}{5}\right]_0^1 - \frac{n+1}{5}\int_0^1 x^n e^{5x}\,\mathrm{d}x$$
$$= \frac{e^5}{5} - \frac{n+1}{5}I_n.$$

6. [0.5 pt] For $0 \leq x \leq 1$, $e^{5x} \geq 1$ and $x^n \geq 0$. So $x^n \leq x^n e^{5x}$; by integration,

$$I_n \geq \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. [0.5 pt] The function $x \mapsto e^{5x}$ is increasing. $[\frac{j}{N}, \frac{j+1}{N}]$, values at the ends framework the function. By integrating and then sleeping,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{5j}{N}} \leq I_0 \leq U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{5j}{N}}.$$

Domestic terms cancel in the difference, so $U_N - L_N = \frac{e^5-1}{N}$.

8. [1 pt] The width is not more than 10^{-2} if $N \geq 100(e^5 - 1)$. The first integer satisfying this bound is $N = 14742$. The following is the calculation of the limits:

```
From math import exp, ceil

def bounds():
    N = ceil(100*(exp(5)-1))
    Low = sum (exp(5*j/N) for j in rows (N))/N
    top = bottom+(exp(5)-1)/N
    return N, low, high
```

Exercise 2 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Like $x > 0$, $f(x) = 0$ equals 4 $+ \ln x = 0$, either $x = e^{-4}$.

4. [1 pt] We're drifting. $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator cancels in $t = e^{-\frac{5}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{9}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{9}{2}$, and the average value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\,\mathrm{d}x = \frac{9}{2(e - 1)}.$$

8. [1 pt] Area $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (3 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $3 + 9t = 10$ gives $t = \frac{7}{9}$; $H = (\frac{34}{9}, \frac{14}{9}, \frac{14}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Therefore

$$d(M, \mathcal{P}) = MH = |\frac{7}{9}| 3 = \frac{7}{3}.$$

4. [1 pt] Lay down $y = z = s$; $x = 10 - 4s$. So $(x, y, z) = (10, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance $\frac{7}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 3$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1| \, \|\vec{n}\| = \frac{2}{3}.$$

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 20 independent identical tests with two exits; $X \sim \mathcal{B}(20, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{20}{2}(\frac{3}{10})^2(\frac{7}{10})^{18} \simeq 0{,}027846$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{20} - 20(\frac{3}{10})(\frac{7}{10})^{19} \simeq 0{,}992363$.

4. [1 pt] E $(X) = 6$, $V(X) = \frac{21}{5}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1 - p)^N$. Probability is therefore 1 $- (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. Calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1 - p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] Events $X = 1$, $X = 2$, $X = 3$ are incompatible, therefore

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{20}{j} (\frac{3}{10})^j (\frac{7}{10})^{20-j} \simeq 0{,}106289.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 20, 3/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 9. The cumulations in rows 8 and 9 are approximately 0.886669 and 0.952038: this confirms the minimality.

White 2 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (0.5 pt)

2. Using a party integration, calculate I_1. (0.5 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for every $n \in \mathbb{N}$. (0.5 pt)

7. We share $[0;1]$ in N intervals of the same length, $N \geq 1$. Write the amounts L_N and U_N of areas of rectangles constructed with values of e^{4x} at left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N ensuring that this framework has a width of not more than 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{3+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^2$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-1) + \ln(x+1) = \ln(63)$ after having specified the set on which this equation is defined. (1 pt)

White 2 · Zero 2 · Continue

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 6$ and point $M(1;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 17 times a two-run event called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 17 events is accepted if the number of hits is between 1 and 3 points Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Corrected · White 2 · Zero 2

Exercise 1 · Comprehensive and follow-up

1. [0.5 pt] The function $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of e^{4x}. Thus $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. [0.5 pt] Choices $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two continuously derivative derivative functions on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Part-by-part integration gives

$$I_1 = \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x$$
$$= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}.$$

3. [0.5 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Like $e^{4x} > 0$, integration on increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The follow-up is therefore positive and decreasing in the broad sense. the same strict: the integrands of I_n and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{4x} \leq e^4$; therefore $0 \leq I_n \leq e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. [0.5 pt] Apply PPI to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{4x}}{4}$. Like $n+1 \geq 1$, the term in $x = 0$ sucks. Where

$$I_{n+1} = \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x$$
$$= \frac{e^4}{4} - \frac{n+1}{4}I_n.$$

6. [0.5 pt] For $0 \leq x \leq 1$, $e^{4x} \geq 1$ and $x^n \geq 0$. So $x^n \leq x^n e^{4x}$; by integration,

$$I_n \geq \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. [0.5 pt] The function $x \mapsto e^{4x}$ is increasing. $[\frac{j}{N}, \frac{j+1}{N}]$, values at the ends framework the function. By integrating and then sleeping,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{4j}{N}} \leq I_0 \leq U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{4j}{N}}.$$

Domestic terms cancel in the difference, so $U_N - L_N = \frac{e^4-1}{N}$.

8. [1 pt] The width is not more than 10^{-2} if $N \geq 100(e^4 - 1)$. The first integer satisfying this bound is $N = 53$. The following is the calculation of the thresholds:

```
From math import exp, ceil

def bounds():
    N = ceil(100*(exp(4)-1))
    Low = sum (exp(4*j/N) for j in rows (N))/N
    top = bottom+(exp(4)-1)/N
    return N, low, high
```

Exercise 2 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3 + \ln x)}{x^2} = \frac{-2 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-2}$ and is

$$f(s) = \frac{3 - 2}{e^{-2}} = e^2.$$

Like $x > 0$, $f(x) = 0$ equals 3 $+ \ln x = 0$, either $x = e^{-3}$.

4. [1 pt] We're drifting. $(-2 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2 - \ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator cancels in $t = e^{-\frac{3}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{7}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{7}{2}$, and the average value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\,\mathrm{d}x = \frac{7}{2(e - 1)}.$$

8. [1 pt] Area $x > 1$; $x^2 - 1 = 63$, therefore $x = 8$ with negative root excluded.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $1 + 9t = 6$ given $t = \frac{5}{9}$; $H = (\frac{14}{9}, \frac{10}{9}, \frac{10}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{5}{9}\vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Therefore

$$d(M, \mathcal{P}) = MH = |\frac{5}{9}| 3 = \frac{5}{3}.$$

4. [1 pt] Lay down $y = z = s$; $x = 6 - 4s$. So $(x, y, z) = (6, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance $\frac{5}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 1$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{5}{9} - 1| \, \|\vec{n}\| = \frac{4}{3}.$$

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 17 independent identical tests with two exits; $X \sim \mathcal{B}(17, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0{,}058110$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0{,}980725$.

4. [1 pt] E $(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1 - p)^N$. Probability is therefore 1 $- (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. Calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 17$. For $j \in \{0, \ldots, n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1 - p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] Events $X = 1$, $X = 2$, $X = 3$ are incompatible, therefore

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{17}{j} (\frac{3}{10})^j (\frac{7}{10})^{17-j} \simeq 0{,}199581.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 17, 3/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 8. The cumulations in rows 7 and 8 are approximately 0.895360 and 0.959723: this confirms the minimality.

White 2 · A

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · An area to share — 4 points

On $[0;1]$, we consider $f(x) = x^2$. We're laying $L_n = \frac{1}{n}\sum_{k=0}^{n-1}\left(\frac{k}{n}\right)^2$ and $U_n = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^2$ for $n \geq 1$.

1. Draw corresponding rectangles for $n = 4$ and justify $L_n \leq \int_0^1 x^2\,\mathrm{d}x \leq U_n$. (1 pt)

2. Calculate $U_n - L_n$ and give a size n which guarantees a width of not more than 10^{-3}. (1 pt)

3. Calculate the integral exactly and compare to the frames. (1 pt)

4. Determine Abscissa $a \in [0;1]$ which divides this area into two equal areas. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{2+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^1$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-2) + \ln(x+2) = \ln(21)$ after having specified the set on which this equation is defined. (1 pt)

White 2 · A · continued

Exercise 3 · Two plans and one projection — 6 points

In an orthonormal coordinate system, $\mathcal{P} : x + y + z = 3$, $\mathcal{Q} : x - y = 0$ and $M(3;0;0)$.

1. Justify that the two planes are sloping. (1 pt)

2. Provide a parametric representation of their intersection d. (1 pt)

3. Determining the Project H of M on d. 2 pts

4. Calculate the distance from M to d. (1 pt)

5. Is this project also that of M on $\mathcal{P}$? Justify. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 18 times a two-run event, called success and failure.
and the probability of success is $p = \frac{1}{5}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently.
Determine the smallest integer N for which the probability of achieving at least one success is
An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 18\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 18 events is accepted if its number of successes is between 1 and 3, limits
Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$.
Run the function and check the minimality of the result. (1 pt)

Corrected · White 2 · A

Exercise 1 · An area to share

1. Over each width interval 1 /n, x^2 is understood between its values at the ends. to the left are minor and those to the right are increasing. $n = 4$, left heights 0, 1 /16, 1/4, 9/16, rights 1/16, 1/4, 9/16, 1.

2. The internal terms shall cancel: $U_n - L_n = 1/n$. Take $n \geq 1000$.

3. $\int_0^1 x^2\,\mathrm{d}x = 1/3$. $n = 4$, $L_4 = 7/32$ and $U_4 = 15/32$ frame well 1/3; width is 1/4.

4. $\int_0^a x^2\,\mathrm{d}x = a^3/3 = 1/6$ given $a = \sqrt[3]{1/2}$. The strict growth of $a \mapsto a^3$ guarantees uniqueness.

Exercise 2 · A logarithmic function

1. [0.5 pt] In 0 $^+$, the numerator tends to $-\infty$ and denominator to 0 $^+$: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2 + \ln x)}{x^2} = \frac{-1 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-1}$, is positive for 0 $< x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-1}$ and is

$$f(s) = \frac{2 - 1}{e^{-1}} = e^1.$$

Like $x > 0$, $f(x) = 0$ equals 2 $+ \ln x = 0$, either $x = e^{-2}$.

4. [1 pt] We're drifting. $(-1 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1 - \ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator cancels in $t = e^{-\frac{1}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 2\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{5}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{5}{2}$, and the average value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\,\mathrm{d}x = \frac{5}{2(e - 1)}.$$

8. [1 pt] Area $x > 2$; $x^2 - 4 = 21$, therefore $x = 5$, the negative root being excluded.

Exercise 3 · Two plans and one projection

1. Normal $(1;1;1)$ and $(1;-1;0)$ non-colonial, so pruning planes.

2. $x=y=t$, $z=3-2t$. Thus $d:(t;t;3-2t)$, Director $u=(1;1;-2)$.

3. $H=(t;t;3-2t)$; $MH=(t-3;t;3-2t)$. Equation $MH\cdot u=6t-9=0$ given $t=3/2$, or $H(3/2;3/2;0)$. A co-ordinated and system: 1 point; resolution and control of membership: 1 point.

4. $MH=\sqrt{(3/2)^2+(3/2)^2}=3/\sqrt{2}$.

5. No: M already belongs to $\mathcal{P}$ and its plans in this area are M, while $H\neq M$.

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 18 independent identical tests with two exits; $X\sim\mathcal{B}(18,\frac{1}{5})$.

2. [0.5 pt] P $(X=2)=\binom{18}{2}(\frac{1}{5})^2(\frac{4}{5})^{16}\simeq 0{,}172263$.

3. [0.5 pt] $1-P(X=0)-P(X=1)=1-(\frac{4}{5})^{18}-18(\frac{1}{5})(\frac{4}{5})^{17}\simeq 0{,}900921$.

4. [1 pt] E $(X)=\frac{18}{5}$, $V(X)=\frac{72}{25}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1-p)^N$. Probability is therefore 1 $-(\frac{4}{5})^N$. It is strictly increasing with N, because $0<\frac{4}{5}<1$. Calculations give

$$P_{13}\simeq 0{,}945024<0{,}95,\qquad P_{14}\simeq 0{,}956020\geq 0{,}95.$$

Thus 14 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n=18$. For $j\in\{0,\dots,n\}$, an ordered result containing j success and $n-j$ Failures a, out of independence, probability $p^j(1-p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X=j)=\binom{n}{j}p^j(1-p)^{n-j},\qquad p=\frac{1}{5}.$$

7. [0.5 pt] Events $X=1$, $X=2$, $X=3$ are incompatible, therefore

$$P(1\leq X\leq 3)=\sum_{j=1}^{3}\binom{18}{j}(\frac{1}{5})^j(\frac{4}{5})^{18-j}\simeq 0{,}483011.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 18, 2/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 7. The cumulations in rows 6 and 7 are approximately 0.948729 and 0.983720: this confirms the minimality.

White 2 · B

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Algebraic area, geometric area — 4 points

It is considered $f(x) = x - 1$ on $[0;3]$.

1. Draw the curve and indicate the sign of f. (1 pt)

2. Calculate $\int_0^3 f(x)\,\mathrm{d}x$, then the geometric area between the curve and the axis of the apses. (1 pt)

3. Determine the mean value of f and build the rectangle of the same algebraic area on $[0;3]$. (1 pt)

4. For $a \in [1;3]$resolve $\int_0^a f(x)\,\mathrm{d}x = 0$ and explain the compensation. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{5+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^4$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-3) + \ln(x+3) = \ln(27)$after having specified the set on which this equation is defined. (1 pt)

White 2 · B · continued

Exercise 3 · A straight line whose position depends on a parameter — 6 points

In an orthonormal coordinate system, $\mathcal{P} : x + 2y - z = 1$. For $m \in \mathbb{R}$, d_m is defined by $(x;y;z) = (0;0;m) + t(1;m;1)$, $t \in \mathbb{R}$.

1. Show that each d_m is a right. (1 pt)

2. Determine the intersection with $\mathcal{P}$ where $m \neq 0$. 2 pts

3. Study separately $m = 0$. (1 pt)

4. For $m = 1$, calculate the point of intersection. (1 pt)

5. The Right d_1 Is she orthogonal to the plan? (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 20 times a two-run event, called success and failure. The tests are independent and the probability of success is $p = \frac{1}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 20 events is accepted if its number of successes is between 1 and 3, bounds Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Corrected · White 2 · B

Exercise 1 · Algebraic area, geometric area

1. Right through $(0;-1)$ and $(1;0)$; negative sign up to 1, positive after.

2. $[x^2/2-x]_0^3 = 3/2$. Negative area in absolute value: 1 $/2$; positive area : 2, therefore geometric area 5/2. Two results at 0.5 points.

3. Average $(3/2)/3 = 1/2$. Rectangle width 3 and height 1/2, so area 3/2.

4. $a^2/2 - a = a(a-2)/2 = 0$; the only solution in $[1;3]$ is $a = 2$. The two triangles on $[0;1]$ and $[1;2]$ have each area 1/2 and opposite signs.

Exercise 2 · A logarithmic function

1. [0.5 pt] In 0 $^+$, the numerator tends to $-\infty$ and denominator to 0 $^+$: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5+\ln x)}{x^2} = \frac{-4-\ln x}{x^2}.$$

The denominator is positive. $s = e^{-4}$, is positive for 0 $< x < s$ and negative for $x > s$. Thus f growth strictly on $]0,s]$, then decreases strictly on $[s,+\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-4}$ and is

$$f(s) = \frac{5-4}{e^{-4}} = e^4.$$

Like $x > 0$, $f(x) = 0$ equals 5 $+\ln x = 0$, either $x = e^{-5}$.

4. [1 pt] We're drifting. $(-4-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4-\ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator cancels in $t = e^{-\frac{7}{2}}$ and goes from negative to positive. The function is concave on $]0,t]$, convex $[t,+\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0,+\infty[$. On $]0,s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0,s[$. On $[s,+\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s,+\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0,+\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{11}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{11}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{11}{2(e-1)}.$$

8. [1 pt] Area $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

Exercise 3 · A straight line whose position depends on a parameter

1. $(1;m;1) \neq 0$ because its first co-ordinate is equal to 1.

2. The substitution gives $t+2mt-(m+t)=1$ or 2 $mt=1+m$. If $m \neq 0, t=(1+m)/(2m)$: unique point $((1+m)/(2m);(1+m)/2;m+(1+m)/(2m))$. Equation 1 point, solution and coordinates 1 point.

3. For $m=0$, equality is $0 \quad =1$: empty intersection, straight strictly parallel to the plane.

4. $t=1$ and point $(1;1;2)$, which checks $1+2-2=1$.

5. No: $(1;1;1)$ is not colinear to normal $(1;2;-1)$. Being stinging does not mean being orthogonal.

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 20 independent identical tests with two exits; $X \sim \mathcal{B}(20, \frac{1}{10})$.

2. [0.5 pt] P $(X=2)=\binom{20}{2}(\frac{1}{10})^2(\frac{9}{10})^{18} \simeq 0{,}285180$.

3. [0.5 pt] $1-P(X=0)-P(X=1)=1-(\frac{9}{10})^{20}-20(\frac{1}{10})(\frac{9}{10})^{19} \simeq 0{,}608253$.

4. [1 pt] E $(X)=2, V(X)=\frac{9}{5}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1-p)^N$. Probability is therefore 1 $-(\frac{9}{10})^N$. It is strictly increasing with N, because $0<\frac{9}{10}<1$. Calculations give

$$P_{28} \simeq 0{,}947665 < 0{,}95, \qquad P_{29} \simeq 0{,}952899 \geq 0{,}95.$$

Thus 29 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n=20$. For $j \in \{0,\ldots,n\}$, an ordered result containing j success and $n-j$ Failures a, out of independence, probability $p^j(1-p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X=j)=\binom{n}{j}p^j(1-p)^{n-j}, \qquad p=\frac{1}{10}.$$

7. [0.5 pt] Events $X=1, X=2, X=3$ are incompatible, therefore

$$P(1 \leq X \leq 3)=\sum_{j=1}^{3}\binom{20}{j}(\frac{1}{10})^j(\frac{9}{10})^{20-j} \simeq 0{,}745470.$$

8. [1 pt]

```
From math import comb

def Quantile():
    n, p = 20, 1/10
    j, cumulation = 0, (1-p)**n
    while cumulation < 0.95:
        j += 1
        cumulation += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulation is 1, so the loop stops. It returns 4. The cumulations in rows 3 and 4 are about 0.867047 and 0.956826: this confirms the minimality.

Final white · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^1$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-2)+\ln(x+2)=\ln(32)$ after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel rotating at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=9-3\cos\left(\frac{\pi t}{5}\right)$ where: t is expressed in seconds and $h(t)$ in metres. We're studying a full tour, $t\in[0;10]$.

1. Give an extended model period to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and to draw up a table of changes in h on $[0;10]$. (0.5 pt)

3. Determine every moment $t\in[0;10]$ such that $h(t)=\frac{21}{2}$. (0.5 pt)

4. Calculate the average value of h on $[0;10]$. (1 pt)

5. Study the convexity of h the study interval and determine the coordinates of the points his curve inflection. (0.5 pt)

6. Determine the total duration during which the height is at least equal to the level of the question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the moments when they are reached. (0.5 pt)

8. Calculate vertical displacement $h(10)-h(0)$, then the sum of the height travelled at the climb explain why these two results are different. (1 pt)

Final white · Zero 1 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 11$ and point $M(2;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 20 times a two-run event, called success and failure. The tests are independent and the probability of success is $p = \frac{1}{5}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. For N independent tests of the same parameter p, we note F_N the frequency of success. Demonstrate expressions of $E(F_N)$ and $V(F_N)$ according to the N and p. (0.5 pt)

8. Using Bienaymé-Chebychev, determine a size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Corrected · Final white · Zero 1

Exercise 1 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2+\ln x)}{x^2} = \frac{-1-\ln x}{x^2}.$$

The denominator is positive. $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-1}$ and is

$$f(s) = \frac{2-1}{e^{-1}} = e^1.$$

Like $x > 0$, $f(x) = 0$ equals $2 + \ln x = 0$, either $x = e^{-2}$.

4. [1 pt] We're drifting. $(-1-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1-\ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator cancels in $t = e^{-\frac{1}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s: the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 2\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{5}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{5}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{5}{2(e-1)}.$$

8. [1 pt] Area $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

Exercise 2 · Periodic phenomenon

1. [0.5 pt] Period 10 s; minimum 6 m, maximum 12 m.

2. [0.5 pt] h $'(t) = (\frac{3}{5}\pi)\sin(\frac{\pi t}{5})$; growth on $[0, 5]$, decline on $[5, 10]$.

3. [0.5 pt] The equation is equivalent to $\cos(\frac{\pi t}{5}) = -\frac{1}{2}$. On a tower, $\frac{\pi t}{5} \in [0, 2\pi]$; the appropriate angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Thus $t = \frac{10}{3}$ s or $t = \frac{20}{3}$ s.

4. [1 pt] A antiderivative is $F(t) = 9t - \frac{15}{\pi}\sin(\frac{\pi t}{5})$. Like $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{10}\int_0^{10} h(t)\,\mathrm{d}t = \frac{F(10) - F(0)}{10} = \frac{90}{10} = 9 \text{ m}.$$

5. [0.5 pt] The second derivative is $h''(t) = \frac{3\pi^2}{25}\cos(\frac{\pi t}{5})$. Its sign is that of the cosine: positive on $[0, \frac{5}{2}[$ and $]\frac{15}{2}, 10]$, negative between these two intervals. The function is convex on $[0, \frac{5}{2}]$ and $[\frac{15}{2}, 10]$Concave on $[\frac{5}{2}, \frac{15}{2}]$. The two changes of convexity give the points of inflection $(\frac{5}{2}, 9)$ and $(\frac{15}{2}, 9)$.

6. [0.5 pt] Between $\frac{10}{3}$ and $\frac{20}{3}$; duration $\frac{10}{3}$ s.

7. [0.5 pt] h ′ Achieved $\frac{3}{5}\pi$ m s^{-1} in $t = \frac{5}{2}$ and its opposite in $t = \frac{15}{2}$.

8. [1 pt] Movement $h(10) - h(0) = 0$. Ascent of 6 m then descent of 6 m: cumulative distance 12 m, equal to the integral of $|h'|$.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $2 + 9t = 11$ gives $t = 1$; $H = (3, 2, 2)$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Therefore

$$d(M, \mathcal{P}) = MH = |1|\,3 = 3.$$

4. [1 pt] Lay down $y = z = s$; $x = 11 - 4s$. So $(x, y, z) = (11, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance 3.

6. [0.5 pt] x $+ 2y + 2z = 2$; same normal vector and passage through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |1 - 1|\,\|\vec{n}\| = 0.$$

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 20 independent identical tests with two exits; $X \sim \mathcal{B}(20, \frac{1}{5})$.

2. [0.5 pt] P $(X = 2) = \binom{20}{2}(\frac{1}{5})^2(\frac{4}{5})^{18} \simeq 0{,}136909$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{20} - 20(\frac{1}{5})(\frac{4}{5})^{19} \simeq 0{,}930825$.

4. [1 pt] E $(X) = 4$, $V(X) = \frac{16}{5}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1 - p)^N$. Probability is therefore 1 $- (\frac{4}{5})^N$. It is strictly increasing with N, because $0 < \frac{4}{5} < 1$. Calculations give

$$P_{13} \simeq 0{,}945024 < 0{,}95, \qquad P_{14} \simeq 0{,}956020 \geq 0{,}95.$$

Thus 14 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 20$. For $j \in \{0, \dots, n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1 - p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{1}{5}.$$

7. [0.5 pt] If $B_1, \ldots, B_N$ are independent indicators of success, $F_N = \frac{B_1+\cdots+B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{1}{5}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{4}{25}}{N}.$$

Independence is required for the addition of variances, not for the linearity of expectation.

8. [1 pt] Bienaymé-Chebychev applied to F_N gives

$$P(|F_N - p| \geq 0{,}02) \leq \frac{p(1-p)}{N(0{,}02)^2}.$$

To ensure that this probability is not more than 0.05, it is sufficient to

$$N \geq \frac{\frac{4}{25}}{0{,}05 \times (0{,}02)^2} = 8000.$$

As a complement, we then have $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Size 8000 is sufficient; inequality being an increase, it does not necessarily provide the minimum size for the probability
That's right.

Final white · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^4$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-3)+\ln(x+3)=\ln(40)$ after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel rotating at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=12-4\cos\left(\frac{\pi t}{4}\right)$ where: t is expressed in seconds and $h(t)$ in metres. We're studying a full tour, $t\in[0;8]$.

1. Give an extended model period to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and to draw up a table of changes in h on $[0;8]$. (0.5 pt)

3. Determine every moment $t\in[0;8]$ such that $h(t)=14$. (0.5 pt)

4. Calculate the average value of h on $[0;8]$. (1 pt)

5. Study the convexity of h the study interval and determine the coordinates of the points his curve inflection. (0.5 pt)

6. Determine the total duration during which the height is at least equal to the level of the question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the moments when they are reached. (0.5 pt)

8. Calculate vertical displacement $h(8)-h(0)$, then the sum of the height travelled at the climb explain why these two results are different. (1 pt)

Final white · Zero 2 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 7$ and point $M(3;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 17 times a two-run event called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. For N independent tests of the same parameter p, we note F_N the frequency of success. Demonstrate expressions of $E(F_N)$ and $V(F_N)$ according to the N and p. (0.5 pt)

8. Using Bienaymé-Chebychev, determine a size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Corrected · Final white · Zero 2

Exercise 1 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5 + \ln x)}{x^2} = \frac{-4 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-4}$ and is

$$f(s) = \frac{5 - 4}{e^{-4}} = e^4.$$

Like $x > 0$, $f(x) = 0$ equals $5 + \ln x = 0$, either $x = e^{-5}$.

4. [1 pt] We're drifting. $(-4 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4 - \ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator cancels in $t = e^{-\frac{7}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s: the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{11}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{11}{2}$, and the average value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\,\mathrm{d}x = \frac{11}{2(e - 1)}.$$

8. [1 pt] Area $x > 3$; $x^2 - 9 = 40$, so $x = 7$, negative root being excluded.

Exercise 2 · Periodic phenomenon

1. [0.5 pt] Period 8 s; minimum 8 m, maximum 16 m.

2. [0.5 pt] h $'(t) = \pi\sin(\frac{\pi t}{4})$; growth on $[0, 4]$, decline on $[4, 8]$.

3. [0.5 pt] The equation is equivalent to $\cos(\frac{\pi t}{4}) = -\frac{1}{2}$. On a tower, $\frac{\pi t}{4} \in [0, 2\pi]$; the appropriate angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Thus $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. [1 pt] A antiderivative is $F(t) = 12t - \frac{16}{\pi}\sin(\frac{\pi t}{4})$. Like $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{8}\int_0^8 h(t)\,\mathrm{d}t = \frac{F(8) - F(0)}{8} = \frac{96}{8} = 12 \text{ m}.$$

5. [0.5 pt] The second derivative is $h''(t) = \frac{4\pi^2}{16}\cos(\frac{\pi t}{4})$. Its sign is that of the cosine: positive on $[0,2[$ and $]6,8]$, negative between these two intervals. The function is convex on $[0,2]$ and $[6,8]$, concave on $[2,6]$. The two changes of convexity give the points of inflection $(2,12)$ and $(6,12)$.

6. [0.5 pt] Between $\frac{8}{3}$ and $\frac{16}{3}$; duration $\frac{8}{3}$ s.

7. [0.5 pt] h ′ reached 1π m s^{-1} in $t = 2$ and its opposite in $t = 6$.

8. [1 pt] Movement $h(8) - h(0) = 0$. Ascent of 8 m then descent of 8 m: cumulative distance 16 m, equal to the total of $|h'|$.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (3 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $3 + 9t = 7$ gives $t = \frac{4}{9}$; $H = (\frac{31}{9}, \frac{8}{9}, \frac{8}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1,2,2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{4}{9}\vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Therefore

$$d(M,\mathcal{P}) = MH = |\frac{4}{9}|\,3 = \frac{4}{3}.$$

4. [1 pt] Lay down $y = z = s$; $x = 7 - 4s$. So $(x, y, z) = (7,0,0) + s(-4,1,1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance $\frac{4}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 3$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4,1,1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N,\mathcal{P}) = d(N,\mathcal{P}\cap\mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{4}{9} - 1|\,\|\vec{n}\| = \frac{5}{3}.$$

Exercise 4 · Repetition and estimation

1. [0.5 pt] Repetition of a fixed number of 17 independent identical tests with two exits; $X \sim \mathcal{B}(17, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0{,}058110$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0{,}980725$.

4. [1 pt] E $(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.

5. [0.5 pt] For N independent tests, the probability of no success is $(1-p)^N$. Probability is therefore 1 $-(\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. Calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest number of suitable tests.

6. [0.5 pt] Note: $n = 17$. For $j \in \{0,\dots,n\}$, an ordered result containing j success and $n - j$ Failures a, out of independence, probability $p^j(1-p)^{n-j}$. There is $\binom{n}{j}$ choice of positions of success. results are incompatible, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] If $B_1, \ldots, B_N$ are independent indicators of success, $F_N = \frac{B_1+\cdots+B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{3}{10}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{21}{100}}{N}.$$

Independence is required for the addition of variances, not for the linearity of expectation.

8. [1 pt] Bienaymé-Chebychev applied to F_N gives

$$P(|F_N - p| \geq 0{,}02) \leq \frac{p(1-p)}{N(0{,}02)^2}.$$

To ensure that this probability is not more than 0.05, it is sufficient to

$$N \geq \frac{\frac{21}{100}}{0{,}05 \times (0{,}02)^2} = 10500.$$

As a complement, we then have $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Size 10500 is sufficient; inequality being an increase, it does not necessarily provide the minimum size for the probability
That's right.

Final white · A

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{4+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^3$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-1)+\ln(x+1)=\ln(15)$ after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · A variable trigonometric area — 4 points

For $a\in[0;\pi/2]$, we're laying $A(a)=\int_0^a \sin x\cos x\,\mathrm{d}x$.

1. Justify that $A(a)$ is a geometric area. (1 pt)

2. Calculate $A(a)$ and $A(\pi/2)$. (1 pt)

3. Show that this total area is divided in two by the abscisse $\pi/4$. (1 pt)

4. Determine the maximum of $x\mapsto\sin x\cos x$ on $[0;\pi/2]$ justifying. (1 pt)

Final white · A · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 8$ and point $M(1;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · A guarantee and its real accuracy — 6 points

We roll n times independently a balanced piece, and F_n is the battery frequency.

1. Calculate $E(F_n)$ and $V(F_n)$. (1 pt)

2. Establish a boundary $P(|F_n - \frac{1}{2}| \geq 0{,}1)$. (1 pt)

3. Determine the smallest size that allows this bound to guarantee a probability at most 0 ,05. (1 pt)

4. For $n =$ Write the exact probability of a deviation at least 0 ,1 as a sum binomial. (1 pt)

5. One student concludes "500 tests are necessary and each additional test improves frequency Discuss these two statements separately. 2 pts

Corrected · Final white · A

Exercise 1 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Like $x > 0$, $f(x) = 0$ equals $4 + \ln x = 0$, either $x = e^{-4}$.

4. [1 pt] We're drifting. $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator cancels in $t = e^{-\frac{5}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{9}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{9}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{9}{2(e-1)}.$$

8. [1 pt] Area $x > 1$; $x^2 - 1 = 15$, so $x = 4$, the negative root being excluded.

Exercise 2 · A variable trigonometric area

1. The two factors are positive or zero over the interval, so the integral measure is below the Curved.

2. A antiderivative is $\frac{1}{2}\sin^2 x$ by bypass; $A(a) = \frac{1}{2}\sin^2 a$ and $A(\pi/2) = 1/2$.

3. $A(\pi/4) = \frac{1}{2}(\sqrt{2}/2)^2 = 1/4$, half of 1 $/2$. Strict growth in sin over this interval ensures Oneness of sharing.

4. The derivative is $\cos^2 x - \sin^2 x = \cos 2x$, positive before $\pi/4$, negative after. Maximum 1 $/2$ in $\pi/4$.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $1 + 9t = 8$ given $t = \frac{7}{9}$; $H = (\frac{16}{9}, \frac{14}{9}, \frac{14}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Therefore

$$d(M, \mathcal{P}) = MH = |\frac{7}{9}| 3 = \frac{7}{3}.$$

4. [1 pt] Lay down $y = z = s$; $x = 8 - 4s$. So $(x, y, z) = (8, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance $\frac{7}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 1$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1| \, \|\vec{n}\| = \frac{2}{3}.$$

Exercise 4 · A guarantee and its real accuracy

1. Average n Bernoulli independent: $E(F_n) = 1/2$, $V(F_n) = 1/(4n)$.

2. Chebychev with $a = 0.1$: probability not more than $1/(4n \times 0{,}01) = 25/n$ (and always at most 1).

3. $25/n \leq 0{,}05 \iff n \geq 500$. Integer first for this guarantee: 500.

4. For $X = 100F_{100} \sim \mathcal{B}(100; 1/2)$, the event is $X \leq 40$ or $X \geq 60$. By symmetry, probability $2\sum_{k=0}^{40} \binom{100}{k} 2^{-100}$.

5. 500 is sufficient according to an increase, not shown to be necessary (1 pt).
frequency of 1/2: after battery then face it is 1/2, after a new battery 2/3 (1 pt).

Final white · B

RESERVED FOR ASSESSMENT

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^2$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-2)+\ln(x+2)=\ln(32)$after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · Study a period without guessing it — 4 points

Set $f(x)=2+\sin(2x)$ on $\mathbb{R}$.

1. Demonstrate that π is a period and that $\pi/2$ is not one. (1 pt)

2. Decreasing changes in f on $[0;\pi]$. (1 pt)

3. Solve $f(x)\geq 5/2$ on $[0;\pi]$. (1 pt)

4. Calculate the average value of f on $[0;\pi]$ and explain the compensation for the oscillation. (1 pt)

Final white · B · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 12$ and the point $M(2;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Stable average or dependent copies — 6 points

A variable X takes values 1 and 3 with probabilities 1 /2. For $n \geq 1$, M_n is the average of n independent copies of X.

1. Calculate $E(X)$ and $V(X)$. (1 pt)

2. Calculate $E(M_n), V(M_n)$ and increase $P(|M_n - 2| \geq 0{,}2)$. (1 pt)

3. Determine the smallest size to guarantee, by this bound, $P(|M_n - 2| < 0{,}2) \geq 0{,}99$. (1 pt)

4. The independent copies are replaced by $X_1 = \cdots = X_n = X$. Calculate the variance of their and the probability of deviation strictly less than 0.2. (1 pt)

5. In the independent model, express M_n by means of a binomial variable, then write a event to calculate exactly this probability. 2 pts

Corrected · Final white · B

Exercise 1 · A logarithmic function

1. [0.5 pt] In 0^+, the numerator tends to $-\infty$ and denominator to 0^+: limit $-\infty$; in $+\infty$limit 0 by comparative growth.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3+\ln x)}{x^2} = \frac{-2-\ln x}{x^2}.$$

The denominator is positive. $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f growth strictly on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is only reached in $s = e^{-2}$ and is

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Like $x > 0$, $f(x) = 0$ equals $3 + \ln x = 0$, either $x = e^{-3}$.

4. [1 pt] We're drifting. $(-2-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2-\ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator cancels in $t = e^{-\frac{3}{2}}$ and goes from negative to positive. The function is concave on $]0, t]$, convex $[t, +\infty[$. It changes convexity to t and $f(t) = \frac{3}{2t}$: the point inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Poisons $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, limit $-\infty$ in 0 and value 2 m in s : the corollary of the VIT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing by 2 m about $0 < m$: exactly a solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there is exactly Two solutions.

6. [0.5 pt] Poisons $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, drawable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; F is a antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{7}{2}$ and $F(1) = 0$. The integral value is therefore $\frac{7}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{7}{2(e-1)}.$$

8. [1 pt] Area $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

Exercise 2 · Study a period without guessing it

1. $f(x+\pi) = 2 + \sin(2x + 2\pi) = f(x)$. But $f(\pi/4) = 3$ and $f(3\pi/4) = 1$; therefore $\pi/2$ is not a period.

2. $f' = 2\cos 2x$. Growth on $[0; \pi/4]$ from 2 to 3, decrease in $[\pi/4; 3\pi/4]$ up to 1, then growth up to 2.

3. $\sin(2x) \geq 1/2$, with 2 $x \in [0; 2\pi]$, therefore $x \in [\pi/12; 5\pi/12]$.

4. A antiderivative is 2 $x - \frac{1}{2}\cos 2x$; integral 2 π, mean 2. Integral sine over a period value 0: excess and deficit around 2 compensate.

Exercise 3 · Right, plan and planned

1. [0.5 pt] $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $2 + 9t = 12$ gives $t = \frac{10}{9}$; $H = (\frac{28}{9}, \frac{20}{9}, \frac{20}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Like $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{10}{9}\vec{n}$, H is the projected orthogonal of M on the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Therefore

$$d(M, \mathcal{P}) = MH = |\frac{10}{9}| 3 = \frac{10}{3}.$$

4. [1 pt] Lay down $y = z = s$; $x = 12 - 4s$. So $(x, y, z) = (12, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, his last two coordinates are equal. H is on the right and $\overrightarrow{MH}$ is orthogonal to any direction of the plane, therefore to this right. Distance $\frac{10}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 2$; same normal vector and passage through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal right as M ; its project on $\mathcal{P}$ is therefore also H. Or. $H \in \mathcal{P} \cap \mathcal{Q}$. For everything P on this right, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{10}{9} - 1| \, \|\vec{n}\| = \frac{1}{3}.$$

Exercise 4 · Stable average or dependent copies

1. $E(X) = 2$; $V(X) = \frac{1}{2}(1 - 2)^2 + \frac{1}{2}(3 - 2)^2 = 1$.

2. $E(M_n) = 2$, $V(M_n) = 1/n$. Chebychev gives a bound 25/ n.

3. Require 25/ $n \leq 0.01$ given $n \geq 2500$. The complementary probability is then at least 0.99.

4. The average is exactly XIts two values have a deviation 1 to the expectation, so the requested probability is 0.

5. If B count values 3, $B \sim \mathcal{B}(n; 1/2)$ and $M_n = 1 + 2B/n$ (1 pt). The event is 0 $,4n < B < 0{,}6n$, let $\lfloor 0{,}4n \rfloor + 1 \leq B \leq \lceil 0{,}6n \rceil - 1$ (1 pt). The binomial probabilities can be summed up on these integers.